



GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT
PATHOLOGIES VIA PROJECTION
NORMALIZATION, AND THE INTRODUCTION OF
IFOPNG

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THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE

DEPARTMENT OF
COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE
SCHOOL OF HUMANITIES AND DIGITAL SCIENCES
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DATE

June 10, 2026

WORD COUNT

≈8500

ACKNOWLEDGMENTS

The author thanks dr. Giacomo Spiegler for supervision and feedback throughout this project.

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Abstract

This thesis addresses the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse due to a structural-numerical pathology where the architectural necessity of weight-chunking, required to avoid an output parameter explosion, forces the model to be produced in hundreds or thousands of sequential chunks before a single projection step can execute. This repetitive chunk-generation loop accumulates an exceptionally large gradient volume within the memory matrix, blowing up the condition number of the projection correlation matrix to infinity and causing catastrophic inversion failures that the standard regularization parameter λ cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce iFOPNG, an inertial variant of Fisher-Orthogonal Projected Natural Gradient Descent (Garg, Kolhe, Peng, & Gopalam, 2026) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split and Permuted MNIST, Split-CIFAR10 and Split-CIFAR100. The empirical findings reveal that projection normalization completely resolves matrix inversion collapse, and that in the hypernetwork setting all normalized projection methods outperform plain SGD, with iFOPNG consistently outperforming

FOPNG across all benchmarks. However, a standalone hypernetwork optimized with Adam and von Oswald’s functional regularizer consistently surpasses all projection variants, indicating that momentum, which projection methods forgo by design, is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. The MNIST (owned by LeCun et al.) and CIFAR (CIFAR-10 and CIFAR-100, owned by Krizhevsky et al.) datasets are public, anonymized datasets licensed for research. These datasets form the basis for the Split-MNIST, Permuted MNIST, Split-CIFAR10, and Split-CIFAR100 benchmarks evaluated in this work. No human or animal data was collected. Data was acquired via the `torchvision.datasets` API.

FIGURES. All figures and plots were created entirely by the author using `matplotlib` and `Weights & Biases`.

CODE. The baseline FOPNG framework was adapted from [Garg et al. \(2026\)](#); all other execution infrastructure was implemented by the author (<https://github.com/kelares/HyperFisher>). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in $\text{\LaTeX}$ using BibTeX. During the preparation of this work, the author(s) used Gemini and Claude in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citations: Google. (2026). *Gemini* [Large language model]. <https://gemini.google.com>; Anthropic. (2026). *Claude* [Large language model]. <https://claude.ai>.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic forgetting* ([McCloskey & Cohen, 1989](#)). When a network optimizes its parameters for an incoming task, gradient updates blindly overwrite the

weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima.

Modern continual learning (CL) literature is conventionally structured into three core algorithmic families: replay-based strategies that leverage episodic data buffers, regularization-based frameworks that penalize parameter deviations, and architectural or parameter-isolation methods (De Lange et al., 2022). Within the meta-learning domain, hypernetworks (Ha, Dai, & Le, 2016) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network’s complete parameter weights conditioned on task embeddings, as formalized by Oswald, Henning, Grewe, and Sacramento (2020). Concurrently, within the field of geometric optimization, gradient projection methods, exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026), explicitly constrain subsequent updates to lie orthogonal to the subspace of parameter directions critical for previous tasks, with the geometry of orthogonality ranging from Euclidean (OGD) to Fisher-Riemannian (FOPNG).

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection uncovers a critical and previously unexamined optimization pathology. Generating a target network’s full weight vector in a single forward pass would require an output layer whose width equals the total number of target network parameters, which is computationally prohibitive for any non-trivial architecture; because every sub-pass executes before any gradient update, their contributions accumulate additively in the shared meta-model parameters, inflating the gradient magnitude proportionally to K . This gradient inflation propagates into the projection algebra: both the gradient columns stored in memory matrix G and the diagonal Fisher estimate $\hat{F}$ grow in scale with K , rendering the regularization term λI negligible relative to the inflated projection matrix $G^\top \hat{F} G$. The condition number of this matrix exceeds 10^{10} , causing catastrophic inversion failures.

Research Objectives and Algorithmic Framework

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system.

To achieve this, we introduce *projection-scoped normalization* (column-wise ℓ_2 orthonormalization of G combined with absolute-maximum scaling

of $\hat{F}$) and iFOPNG, an inertial FOPNG variant using the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$. These contributions enable stable gradient projection in chunked hypernetworks, with direct applicability to on-device sequential learning where retaining adaptive momentum is critical.

Research Questions

To systematically navigate these boundaries, this thesis addresses the following primary research question:

What numerical pathologies arise when gradient projection methods are applied to chunked hypernetwork architectures, and do modifications to the Fisher metric construction—specifically an inertial combined metric and a monotonic accumulation operator—yield principled improvements over standard projection baselines in continual learning?

This is decomposed into four sub-research questions (Sub-RQs):

- Sub-RQ1 (Synergy vs. Conflict):** Does the joint integration of hard gradient projection with the hypernetwork’s soft functional regularizer yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?
- Sub-RQ2 (Gradient Pathology):** How does the architectural necessity of spawning K sequential weight-chunks inflate the gradient and Fisher magnitudes accumulated before a single projection step, and does rescaling both components independently within the projection algebra suffice to prevent matrix inversion collapse?
- Sub-RQ3 (Parameter Inertia):** Does replacing FOPNG’s task-specific metric $\hat{F}_{\text{new}}$ with the combined inertial metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ yield consistent retention improvements across benchmarks, and does the magnitude of any advantage scale with inter-task Fisher overlap?
- Sub-RQ4 (Fisher Accumulation):** Does accumulating $\hat{F}_{\text{old}}$ via a non-decaying element-wise maximum operator maintain long-horizon retention more effectively than an exponential moving average, and does the monotonic growth of the maximum operator impose a measurable plasticity cost over extended task sequences?

Summary of Empirical Findings

Projection-scoped normalization fully resolves the matrix inversion pathology, recovering stable projection performance across all seeds. The inertial

combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ consistently improves retention over FOPNG, with the advantage scaling monotonically with inter-task Fisher overlap. The MAX accumulation operator outperforms EMA on long task sequences and requires no additional hyperparameters. In the standard-capacity hypernetwork setting, iFOPNG surpasses plain SGD but falls short of Adam; under extreme compression the gap to Adam widens to 20–40 pp and most projection methods drop below SGD entirely. Two structural limitations bound the current approach: the loss of adaptive momentum at each projection step, which cannot be recovered without projection-compatible optimizers; and the heuristic nature of the normalization scheme, which stabilises the projection algebra by discarding the curvature structure that a geometrically principled rescaling would preserve.

3 RELATED WORK

This section builds the theoretical and empirical context for the thesis, tracing a path from foundational continual learning challenges through regularization and projection methods, to hypernetwork architectures and the structural gap at their intersection.

3.1 *Continual Learning and the Stability-Plasticity Dilemma*

Modern machine learning systems are predominantly trained on fixed datasets drawn from a single stationary distribution — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (Grossberg, 1980): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations. When standard architectures trained by empirical risk minimization are exposed to sequential data, they suffer *catastrophic forgetting* (McCloskey & Cohen, 1989) — abrupt and near-total loss of previously learned capabilities upon exposure to new task data.

The continual learning literature addresses this across three canonical scenarios (De Lange et al., 2022): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 Evaluation Protocols: Multi-Head and Single-Head Designs

Within the task-incremental setting, two evaluation protocols govern how the network’s output layer is organized. In the *multi-head* protocol, each task is assigned a dedicated output head selected at test time using the known task identity, eliminating output-level interference between tasks. In the *single-head* protocol, all tasks share one output layer, requiring the backbone alone to resolve inter-task interference — a substantially harder problem. Farquhar and Gal (2018) demonstrated that multi-head evaluations systematically inflate apparent method performance by removing a major source of forgetting, and argued that single-head evaluation is the more rigorous protocol.

This thesis uses multi-head benchmarks for all Split-CIFAR experiments to match the evaluation conditions of Garg et al. (2026) and enable direct numerical comparison. For Split-MNIST, both protocols are evaluated: the multi-head variant replicates Garg et al.’s conditions, while the single-head variant follows Farquhar and Gal (2018)’s recommendation.

It should be noted that Garg et al. (2026) label their Split-MNIST evaluation as single-head; however, replication of their stated first-task protocol — SGD at $\eta = 10^{-5}$ for five epochs — yields approximately 21% first-task accuracy, barely above the random baseline. Replication under a corrected first-task learning rate of $\eta = 10^{-3}$ and the task-incremental single-head protocol — a shared two-neuron output head with task identity provided to the evaluation harness at test time — yields iFOPNG at 80.0% and FOPNG at 78.7%, consistent with the range reported by Garg et al. (2026). The stated rate of $\eta = 10^{-5}$ is therefore an error in the original paper’s hyperparameter table; the protocol itself is correctly identified as task-incremental single-head.

3.3 Regularization-Based Approaches

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) is the canonical representative. EWC estimates the structural importance of each weight using the diagonal of the empirical Fisher Information Matrix $\hat{F}$, and applies a soft quadratic penalty that resists changes to parameters important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2. \quad (1)$$

EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods

are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (De Lange et al., 2022).

3.4 Gradient Projection Methods

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (Farajtabar, Azizan, Mott, & Li, 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K]$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\text{Col}(G)$ before each update:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g. \quad (2)$$

This provides a hard guarantee: the update cannot increase the loss on previously seen tasks. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (Bennani, Doan, & Sugiyama, 2020).

The mathematical appeal of projection methods over EWC lies precisely in this hardness: where EWC applies a soft penalty that can be overwhelmed, projection methods impose a geometric constraint that is either satisfied or not. However, standard Euclidean projection treats the parameter space as flat and assigns equal importance to all directions. Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026) addresses this by performing the projection in the Riemannian geometry defined by the Fisher metric, so that the orthogonality constraint respects the curvature of the loss surface. The related Fisher Natural Gradient (FNG) (Garg et al., 2026) applies the natural gradient without projection, and Orthogonal Natural Gradient (ONG) (Yadav, Mendoza, & Korrapati, 2025) combines natural gradient preconditioning with Euclidean projection — sitting structurally between OGD and FOPNG.

3.5 Hypernetworks for Continual Learning

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (Ha et al., 2016): a meta-model $\mathcal{H}(\cdot; \phi)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t; \phi)$ of a target network, conditioned on a task embedding e_t . Each task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through

the generative bottleneck and impressive parameter compression ratios relative to storing separate networks per task. Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so Oswald et al. (2020) introduced *weight chunking*: the generator iterates over a secondary chunk embedding c_k , producing C_s target parameters per pass and assembling the full weight vector by concatenation. To prevent the hypernetwork from forgetting how to generate prior task configurations, a soft *functional regularizer* is applied in the output space of the generator:

$$\mathcal{L}_{\text{reg}} = \beta \sum_{s=1}^{t-1} \|\mathcal{H}(\phi, e_s) - \theta_s^*\|_2^2, \quad (3)$$

where θ_s^* denotes the weight snapshot saved after task s . This penalty can be interpreted as EWC applied at the meta level: rather than anchoring individual network parameters, it anchors the generated weight configurations for each previously seen task, penalizing drift in the space where task-relevant quantities actually live. Recent extensions have refined the hypernetwork approach: Partial Hypernetworks (Hemati, Lomonaco, Bacciu, & Borth, 2023) reduce the computational overhead of full weight generation, and Prototype Augmented Hypernetworks (Fuente et al., 2025) augment task embeddings with prototype representations to further stabilize sequential learning.

3.6 Identified Gaps and Motivating Pathologies

Whether hard geometric projection constraints are compatible with the soft functional regularization already employed by hypernetworks, or whether layering both simultaneously over-constrains the generator’s plasticity, has not been investigated (Sub-RQ1).

The architectural necessity of weight-chunking introduces a gradient accumulation pathology — inflated condition numbers in the projection kernel — that is unique to the hypernetwork and projection methods combination and has no prior treatment in the literature (Sub-RQ2).

Existing projection methods enforce a rigid subspace boundary on historical gradient directions; whether replacing this boundary with a combined Fisher metric encoding both old and new task curvature yields consistent retention improvements across benchmarks has not been examined (Sub-RQ3).

The historical Fisher estimate underlying Fisher-based projection depends on a choice of accumulation operator; whether a monotonic maximum or a decaying exponential moving average better preserves long-horizon retention has not been directly compared (Sub-RQ4).

These four gaps jointly motivate the sub-research questions stated in Section 2 and the experimental programme described in Section 4.6.

4 METHODS

This section specifies the precise formulation of every method and architectural component used in our experiments, providing sufficient detail for reproducibility. Conceptual motivation and positioning relative to prior work are covered in Section 3; we restrict our attention here to the exact algorithmic instantiations evaluated.

4.1 Continual Learning Protocol

4.1.1 Task Setup and Evaluation Metrics

A continual learning agent receives a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T$ presented in order. At task t , only the data from $\mathcal{T}_t$ is available; prior task data is not retained. After each task is trained, the model is evaluated on all tasks seen so far. We report two aggregate metrics: *average accuracy*

$$A_t = \frac{1}{t} \sum_{i=1}^t a_{t,i}, \quad (4)$$

and *backward transfer*

$$\text{BWT} = \frac{1}{T-1} \sum_{t=2}^T (a_{T,t} - a_{t,t}), \quad (5)$$

where $a_{i,j}$ denotes accuracy on task j immediately after training task i . Negative BWT quantifies catastrophic forgetting; a value of zero indicates perfect retention.

Reporting both metrics jointly is necessary because each can mask failure modes invisible to the other. A method that applies an overly conservative projection — refusing gradient components that would encode new tasks — can achieve near-zero BWT while gaining little average accuracy; conversely, a method that fully retrains on each new task may sustain competitive average accuracy on the most recently seen tasks while accumulating large negative BWT on earlier ones. Only the combination of high accuracy with near-zero BWT constitutes genuine continual learning, distinguishing methods that balance plasticity and stability from those that sacrifice one for the other.

4.1.2 Multi-Head and Single-Head Evaluation

Following Farquhar and Gal (2018), who demonstrated that multi-head evaluations can inflate apparent method performance, we report results under both protocols for MNIST-based benchmarks.

4.2 Fisher Information Matrix and Regularization-Based Continual Learning

4.2.1 Fisher Information Matrix

All Fisher-based methods in this thesis use the *diagonal empirical approximation*

$$\hat{F}_i \approx \frac{1}{N} \sum_{n=1}^N \left(\frac{\partial \mathcal{L}^{(n)}}{\partial \theta_i} \right)^2, \quad (6)$$

estimated over N held-out samples at the end of each task. The full FIM is intractable for our network sizes; the diagonal approximation is standard across all evaluated methods (Garg et al., 2026; Kirkpatrick et al., 2017).

4.2.2 Elastic Weight Consolidation

EWC (Kirkpatrick et al., 2017) augments the task loss with a Fisher-weighted quadratic penalty:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2, \quad (7)$$

where θ_{t-1}^* denotes the parameter snapshot after task $t-1$. EWC operates exclusively through parameter-space penalties and maintains no gradient subspace; it is included as a regularization baseline employing a different mechanism from projection. We use $\lambda = 10^{-3}$ for all hypernetwork configurations and reproduce the per-benchmark values from Garg et al. (2026) Table 1 for standalone evaluations. Fisher estimates are computed over the full training set for MNIST-based benchmarks and over $N=1024$ samples for CIFAR benchmarks.

4.3 Gradient Projection Methods

This subsection presents the full suite of gradient-based methods evaluated in this thesis. SGD and FNG establish unconstrained lower bounds with no forgetting-protection mechanism; OGD through iFOPNG form the projection family that is the primary focus of this work.

4.3.1 Stochastic Gradient Descent

SGD applies vanilla gradient descent with no forgetting protection:

$$\Delta\theta_{\text{SGD}} = -\eta \nabla_{\theta} \mathcal{L}_t. \quad (8)$$

Adam (?) is included as an adaptive-rate variant. Both define the lower bound on retention performance; any continual learning method should improve upon SGD on the forgetting axis.

4.3.2 Natural Gradient and Fisher Natural Gradient

Natural gradient descent (Amari, 1998) preconditions the gradient with the inverse FIM, respecting the curvature of the statistical manifold:

$$\Delta\theta_{\text{nat}} = -\eta \hat{F}^{-1} \nabla_{\theta} \mathcal{L}. \quad (9)$$

With a diagonal Fisher approximation, inversion reduces to element-wise scaling and remains computationally tractable.

FNG (Garg et al., 2026) applies a trust-region constraint to this Riemannian step:

$$v_{\text{FNG}}^* = -\eta \frac{\hat{F}^{-1} g}{\sqrt{g^{\top} \hat{F}^{-1} g}}. \quad (10)$$

FNG imposes no projection subspace and constitutes no continual learning mechanism; it is included to isolate the contribution of Fisher preconditioning from that of gradient projection.

4.3.3 Orthogonal Gradient Descent

OGD (Farajtabar et al., 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of past task gradient directions and updates parameters via the current gradient projected onto the orthogonal complement of $\text{Col}(G)$:

$$\tilde{g} = g - G(G^{\top} G)^{-1} G^{\top} g, \quad \Delta\theta_{\text{OGD}} = -\eta \tilde{g}. \quad (11)$$

OGD operates in Euclidean space, treating all parameter directions symmetrically.

GRADIENT MEMORY MANAGEMENT. In all projection methods, G is maintained as a fixed-budget basis of at most M columns. Garg et al. (2026) set $M = T \times K$ across all benchmarks, so compression never triggers in their experiments. We address this unexamined regime explicitly: incoming gradient batches are appended directly to G ; when the column count exceeds M , the full basis is recompressed via truncated SVD and $G \leftarrow U_{:,1:M}$, retaining the M directions of greatest variance in the accumulated

gradient memory. When projection-scoped normalization is enabled, QR decomposition is additionally applied to each incoming batch at insertion time (Section 4.5.1).

4.3.4 Orthogonal Natural Gradient

ONG applies OGD-style projection to the natural gradient rather than to the raw gradient, combining Fisher preconditioning with a Euclidean orthogonality constraint:

$$\Delta\theta_{\text{ONG}} = -\eta \left(\hat{F}^{-1}g - G(G^\top G)^{-1}G^\top \hat{F}^{-1}g \right). \quad (12)$$

Unlike FOPNG, the projection is measured in Euclidean rather than Fisher metric space, so the orthogonality guarantee does not extend to the statistical manifold. ONG uses the same memory budget as OGD.

4.3.5 Fisher-Orthogonal Projected Natural Gradient Descent

FOPNG (Garg et al., 2026) executes projection in the Riemannian metric defined by the Fisher:

$$v_{\text{FOPNG}}^* = -\eta \frac{\hat{F}^{-1}Pg}{\sqrt{(Pg)^\top \hat{F}^{-1}Pg}}, \quad P = I - \hat{F}G \left(G^\top \hat{F}G + \lambda I \right)^{-1} G^\top, \quad (13)$$

where λ is added to the diagonal for numerical stability. We reproduce the per-benchmark learning rates and λ values from Garg et al. (2026) Table 1 for standalone configurations, and use a uniform $\text{lr}=10^{-3}$, $\lambda=10^{-3}$ for all hypernetwork configurations.

4.3.6 Inertial FOPNG

In FOPNG, the parameter update takes the form

$$v = c \hat{F}_{\text{new}}^{-1}(g - u), \quad (14)$$

where $\hat{F}_{\text{new}}$ serves as the Riemannian metric: both the trust-region constraint $\|v\|_{\hat{F}_{\text{new}}} = \eta$ and the projection objective $\|g - \hat{F}_{\text{new}}^{-1}u\|_{\hat{F}_{\text{new}}}^2$ are measured in the current task’s Fisher geometry. This is natural when no prior tasks exist, but creates an asymmetry once history accumulates: the constraint subspace $\text{Col}(G)$ is built from gradient directions collected under $\hat{F}_{\text{old}}$, yet the projection metric assigns zero curvature to parameters that were important in previous tasks but uninformative on the current one.

iFOPNG is introduced in this thesis as a variant of FOPNG that resolves this asymmetry by adopting the *combined metric*

$$F_c := \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \quad (15)$$

where $\hat{F}_{\text{old}}$ is maintained as the element-wise maximum of all per-task Fisher diagonals accumulated so far. Replacing (14) with $v = c F_c^{-1}(g - u)$, the constrained trust-region update becomes

$$v^* = -\eta \frac{F_c^{-1} P g}{\sqrt{(P g)^\top F_c^{-1} P g}}, \quad P = I - \hat{F}_{\text{old}} G A^{-1} G^\top \hat{F}_{\text{old}}, \quad (16)$$

where $A = G^\top \hat{F}_{\text{old}} F_c^{-1} \hat{F}_{\text{old}} G$. A formal derivation is provided in Appendix A.

The combined metric F_c assigns each coordinate i an effective mass $(\hat{F}_{\text{new}})_i + (\hat{F}_{\text{old}})_i$, suppressing the natural-gradient step size in directions important to any prior task. We call this property *parameter inertia*: accumulated Fisher importance makes a coordinate harder to displace without any explicit penalty term. In direct contrast to EWC (Kirkpatrick et al., 2017), which pulls parameters toward a stored reference point θ^* , iFOPNG maintains no reference point: a parameter that mattered before is not pulled back, it is simply made harder to displace.

4.4 Hypernetwork Architecture

4.4.1 Chunked Hypernetworks

A hypernetwork (Ha et al., 2016) is a meta-model $\mathcal{H}(\cdot; \phi)$ that generates the complete weight vector $\theta_t = \mathcal{H}(e_t; \phi)$ of a target network conditioned on a task embedding e_t . Generating θ_t in a single forward pass would require an output layer of dimension $|\theta_t|$, which is computationally prohibitive for any non-trivial target architecture. *Weight chunking* (Oswald et al., 2020) resolves this by iterating over a secondary chunk embedding c_k , $k = 1, \dots, K$, and producing a block of C_s target parameters per pass; the full weight vector is assembled by concatenation. Chunking makes the hypernetwork parameter-efficient but introduces the chunk-induced gradient accumulation pathology addressed in Section 4.5.1.

4.4.2 Functional Regularizer

The functional regularizer of Oswald et al. (2020) is active for all hypernetwork configurations, with regularization strength β treated as a tunable hyperparameter (see Appendix 8).

4.4.3 Target Network and Dropout

Standalone configurations reproduce the target network architecture of Garg et al. (2026) without modification, including two Dropout($p=0.5$) layers in the SimpleCIFARCNN classifier head. Dropout is removed from the tar-

get network in all hypernetwork configurations: PyTorch’s `functional_call` propagates the parent module’s `.training` mode to the target submodule, causing stochastic masking across all chunk-generation passes during training. The generative bottleneck ($\phi : d_t + d_c \rightarrow d_h \rightarrow C_s$) and the functional regularizer $\mathcal{L}_{\text{reg}}$ provide sufficient implicit regularization in its place.

4.5 Novel Algorithmic Contributions

4.5.1 Projection-Scoped Normalization

When a hypernetwork spawns K sequential weight-chunks before a single projection step executes, the accumulated gradient vector and the diagonal Fisher entries grow in magnitude proportionally to K , inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing the matrix inversion in Equation (13) to fail catastrophically for any practically sized target network.

We resolve this with two independent rescaling operations that address distinct sources of ill-conditioning:

1. **Gradient basis orthonormalization.** Each batch of K gradient vectors collected after task t is replaced by the orthonormal factor Q from its thin QR decomposition before insertion into G :

$$[g_{t,1} \mid \cdots \mid g_{t,K}] \xrightarrow{\text{QR}} Q, \quad Q^\top Q = I. \quad (17)$$

QR is chosen over SVD at insertion time because the operation requires only orthonormalization of the incoming batch — the number of columns is preserved, not reduced, so the ordered factorization that SVD provides is unnecessary overhead. SVD is reserved for the overflow handler: when the column count of G exceeds the budget M , the full basis is recompressed as $G \leftarrow U_{:,1:M}$, retaining the M left singular vectors of greatest variance and discarding the rest. This operation applies *at storage time* and affects only the basis G ; the gradient g entering the projection update itself is the unmodified parameter gradient from the model.

2. **Metric-tensor scaling.** Within each projection computation, all Fisher quantities are jointly divided by a single scale factor derived from the maximum entry of the combined metric:

$$s = \max_i (F_c)_i, \quad \hat{f} = F / s \text{ for } F \in \{F_{\text{old}}, \hat{F}_{\text{new}}, F_c\}, \quad \hat{\lambda} = \lambda / s. \quad (18)$$

Critically, the damping coefficient λ is rescaled by the same factor, so its ratio to the Fisher magnitude is preserved across tasks regardless

of how the Fisher scale drifts as more tasks accumulate. Because the projection kernel A , its pseudo-inverse, and the trust-region denominator are all computed under the same rescaled metric, the direction of v^* is invariant to s ; only the condition number of A is reduced, which is the sole purpose of the operation.

Together, these operations ensure $G^\top \tilde{F}G$ remains well-conditioned regardless of the number of chunks spawned per task. Gradient orthonormalization eliminates column-scale inflation in G ; metric-tensor normalization bounds the Fisher entries to $[0, 1]$, preventing the kernel from becoming numerically singular. Subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation alters the geometry of the projection.

4.6 Experimental Design

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ3 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and iFOPNG against FOPNG respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ4 requires a long-horizon comparison of two Fisher accumulation operators on the same elastic preconditioner. Hyperparameter choices fall into two categories: those inherited directly from the literature, which require no further justification, and those specific to the novel hypernetwork experiments, each of which is justified below.

4.6.1 Hyperparameter Provenance

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from Garg et al. (2026) Table 1. iFOPNG uses the same values as FOPNG, since the inertial combined Fisher introduces no hyperparameters beyond those FOPNG already defines. ONG (Yadav et al., 2025) is treated as a composite method: it applies OGD’s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for OGD and FNG; its values therefore follow OGD’s Table 1 entries.

DEVIATION FOR FIRST-TASK INITIALIZATION. We deviated from the original protocol in one specific instance: first-task initialization for MNIST-based benchmarks. While Garg et al. (2026) state they used the method’s

own learning rate for first-task SGD training (e.g., $\eta = 10^{-5}$ for FOPNG), our empirical replication of this protocol yielded an initial accuracy of only $\approx 21\%$. As demonstrated by the learning rate convergence sweep detailed in Appendix C (Figure 15), this published rate falls well below the empirical convergence threshold, effectively failing to establish a functional baseline. To ensure a clean initial projection subspace, we standardized the first-task SGD learning rate to 10^{-3} for MNIST-based benchmarks and 10^{-2} for CIFAR-based benchmarks. Beyond this necessary empirical correction, no tuning was performed on the published values.

HYPERNETWORK EXPERIMENTS. Settings not covered by Garg et al. (2026) follow the canonical hypernetwork framework of Oswald et al. (2020); exact values are detailed in Appendix 8. Because computing a full Fisher estimate requires a complete K -chunk generation pass per training sample, estimation is capped at 1,024 samples to maintain computational tractability — a standard approximation consistent with prior CIFAR-10 evaluations (Kirkpatrick et al., 2017). The gradient budget remains $K = 80$ per task, with memory compression handled via QR-and-SVD (Section 4.3). First-task training uses AdamW to seed the memory matrix G and Fisher estimate $\hat{F}$ from a clean geometric basis.

UNIFORM LEARNING RATE CONTROL. To isolate the projection constraint as the sole independent variable in the hypernetwork experiments, all methods use a uniform learning rate of 10^{-3} rather than method-specific rates. This prevents convergence-speed artifacts from conflating the structural effects of the projection geometry.

4.6.2 Sub-RQ1 — Constructive Synergy or Architectural Conflict

EXPERIMENTAL DESIGN. Sub-RQ1 investigates whether layering a hard geometric projection on top of a hypernetwork’s soft functional regularizer yields a net benefit, or if the loss of adaptive momentum makes a standard Adam-regularized hypernetwork superior. To separate architectural conflicts from benchmark-specific artifacts, we evaluate across three datasets at varying levels of compression, comparing an unconstrained standalone network (Panel a) against a hypernetwork (Panel b).

BENCHMARK SELECTION. We utilize a three-benchmark progression. **Benchmark 1 (Split-MNIST)** utilizes a deliberately “suffocated” hypernetwork (generator hidden dimension $d_h = 8$) to expose differential effects without hitting the near-perfect accuracy ceilings observed at higher capacities ((see Appendix 8, Figure 11)). To ensure convergence under these constraints, each task is trained for 15 epochs with a batch size of 64.

Benchmark 2 (Split-CIFAR10) serves as the primary test, offering genuine inter-task interference and utilizing a standard-capacity hypernetwork trained for 50 epochs. **Benchmark 3 (Split-CIFAR100)** extends this sequence to 10 tasks to test scalability. Comparing performance within and across these panels reveals whether projection methods universally dominate, conditionally succeed based on compression, or systematically fail due to momentum stripping.

4.6.3 Sub-RQ2 — Projection-Scoped Normalization

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. iFOPNG is run on the Split-CIFAR10 hypernetwork (Benchmark 2) under three normalization conditions, all other hyperparameters held constant:

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With several hundred chunks accumulated per forward pass, the gradient columns of G and the entries of $\hat{F}$ are expected to reach extreme magnitudes, inflating the condition number of the projection kernel and driving the matrix inversion to NaN. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G ; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are scaled by the maximum entry of the metric tensor F_c within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number of the projection kernel A is logged before each inversion. A condition number that diverges

in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings and the uniform learning rate of 10^{-3} , so any performance difference is attributable to normalization alone.

4.6.4 Sub-RQ3 — Does iFOPNG’s Elastic Preconditioner Help

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ3 asks whether the elastic combined Fisher $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ provides a consistent retention improvement over rigid FOPNG, and whether any advantage is robust to the evaluation protocol.

BENCHMARK SELECTION AND RATIONALE. To test the robustness of the elastic preconditioner, the full method suite is evaluated across four standalone benchmarks. **Permuted-MNIST** tests long-horizon retention under high inter-task Fisher overlap. **Split-MNIST** is evaluated under both multi-head (Garg et al., 2026) and single-head (Farquhar & Gal, 2018) protocols to determine if the iFOPNG advantage survives output-level interference. Finally, **Split-CIFAR10** and **Split-CIFAR100 (multi-head)** provide the most discriminative environments with stronger inter-task interference. Hypernetwork configurations for Split-MNIST and Split-CIFAR10 are also included for completeness to verify whether the elastic advantage survives extreme parameter compression. Comparing iFOPNG against FOPNG across this suite isolates the specific retention benefit of the elastic preconditioner, while comparisons against OGD verify the necessity of the Fisher metric itself.

4.6.5 Sub-RQ4 — Fisher Accumulation Operator

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ4 holds the elastic formulation of Sub-RQ3 fixed and isolates the operator used to accumulate the historical Fisher F_{old} . The non-decaying maximum $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ guarantees that any parameter ever deemed important to a past task retains that importance permanently. This is protective for early tasks but monotonic by construction: the elastic inertia boundary can only tighten, never relax, so in a sufficiently long sequence the accumulated metric may immobilise the parameters needed to learn later tasks.

EXPERIMENTAL DESIGN. iFOPNG is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1 - \alpha) F_{\text{old}} + \alpha F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy. If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ4 is answered in full.

5 RESULTS

This section reports the empirical outcomes of the four sub-research questions in the order stated in the introduction. For each, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns; detailed interpretation is reserved for the Discussion. *Average accuracy* (A_T) is the primary evaluation metric, and *backward transfer* (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix B (Figures 7–9).

5.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

This subsection examines whether combining gradient projection with the hypernetwork’s functional regularizer yields better continual learning than either component alone. Three hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network, an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$), and a ten-task Split-CIFAR100 network.

STANDARD HYPERNETWORK — SPLIT-CIFAR10. Figure 1 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled (see Section 5.2). Plain SGD at 84.1% establishes the unconstrained lower reference point, and Adam at 90.1% defines the upper frontier a standalone hypernetwork can reach without projection. Every projection method surpasses SGD: iFOPNG leads at 85.6%, followed by EWC (81.5%), FNG (80.9%), OGD

(78.3%), FOPNG (77.5%), and ONG (67.2%), confirming a partial constructive synergy with the hypernetwork’s regularizer. However, none match Adam, which leads by 4.5 pp over the best projection method.

The BWT values reveal an instructive asymmetry: Adam has effectively zero backward transfer ($\text{BWT} \approx 0.000$), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013 ; FOPNG: -0.015).

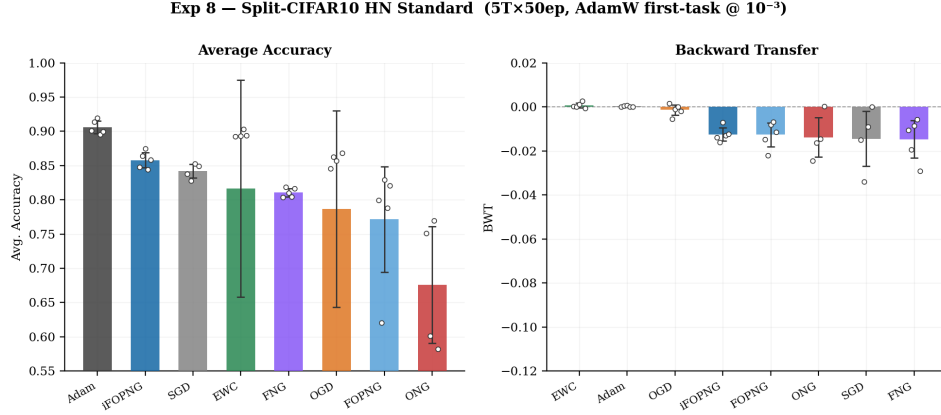


Figure 1: **Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3})**. Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. Adam defines the projection-free upper frontier; SGD defines the unconstrained lower reference. All projection methods trained with full normalization (Section 5.2) exceed SGD but fall short of Adam.

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$. The embedding dimension $d_h = 8$ was selected via a three-point sweep over $d_h \in \{4, 8, 16\}$, presented as an ablation in Figure 11. That sweep shows the projection–Adam gap is largest at $d_h = 8$ (23.5 pp), falls to 14.0 pp at $d_h = 4$ where seed variance becomes extreme and results near-random, and shrinks to 9.4 pp at $d_h = 16$ as the projection methods partially recover. $d_h = 8$ was retained as the most diagnostically informative level: the failure mode is severe and replicable, but the network has sufficient capacity for the projections to converge consistently.

Figure 2 presents the full method comparison at $d_h = 8$. Adam leads at 93.4%. The Fisher-free projection method OGD is the strongest projection baseline at 86.6%, followed by EWC at 84.5% and SGD at 79.8%. The Fisher-based methods fall sharply below: iFOPNG at 69.9%, FOPNG at 59.9%, and FNG at 54.8% — gaps of 23–39 pp against Adam.

The BWT panel reveals the mechanism underlying this collapse. iFOPNG, FOPNG, and FNG all achieve near-zero backward transfer ($\text{BWT} \approx 0.00$), while

Adam has small negative BWT (-0.015) and SGD forgets more substantially (-0.13).

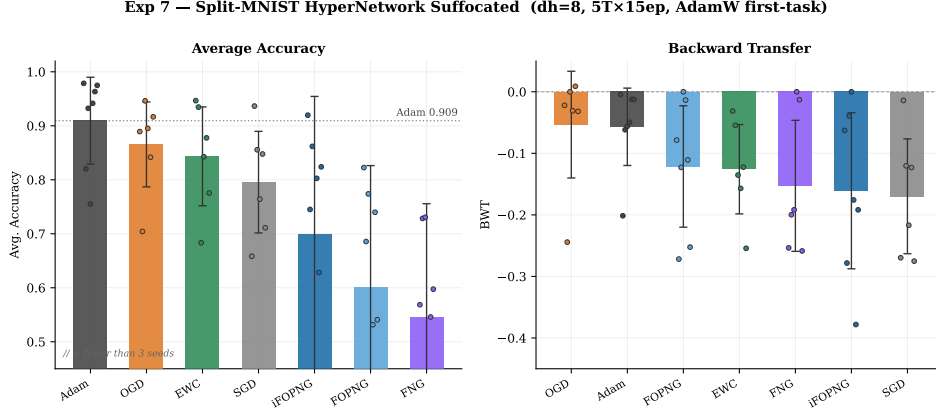


Figure 2: **Split-MNIST SH hypernetwork, suffocated regime ($d_h = 8$, 5 tasks, 15 epochs per task, AdamW first-task initialisation)**. Left: average accuracy (bars show mean over seeds; circles show individual seeds; note that several methods completed fewer than three seeds). Right: backward transfer. The dotted line marks Adam’s mean accuracy (93.4%). Fisher-based projection methods (iFOPNG, FOPNG, FNG) achieve near-zero BWT but much lower accuracy than Adam, revealing a plasticity failure rather than a retention failure in the extreme compression regime. The Fisher-free OGD is the best-performing projection method at 86.6%.

EXTENDED TASK COUNT — SPLIT-CIFAR100. At ten tasks on Split-CIFAR100 (Appendix Figure 14), EWC unexpectedly leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% (random baseline: 10.0%). This partial rank reversal, in which soft regularization overtakes both Adam and gradient projection, differs from the five-task CIFAR10 pattern and is addressed in the Discussion.

5.2 Sub-RQ2: Normalization Resolves the Gradient Pathology

This subsection reports the three-condition normalization ablation (Split-CIFAR10 hypernetwork; 197 chunks, ≈ 6000 output neurons; unnormalized conditions ran for 10 epochs per task with early stopping, full normalization for 50 epochs). Figure 3 traces the $\log_{10}$ condition number of the projection kernel $G^\top \hat{F}G$ over training steps; Figure 4 shows the resulting average accuracy and BWT under each condition.

WITHOUT NORMALIZATION. The right panel of Figure 3 shows the condition number reaching $10^{10.8}$ at the T2→T3 task transition for the most-affected seed, stabilizing near $10^{8.8}$ – $10^{9.0}$ for two others. At this scale,

numerical inversion of the projection kernel is unreliable, effectively zeroing the projected gradient so that the method operates as an unconstrained optimizer with unstable Fisher scaling. The result is 72.2% average accuracy and $\text{BWT} = -0.070$ — worse than plain SGD (84.1%) despite identical hyperparameters. The pathology therefore not only eliminates the benefit of projection but actively degrades performance below the projection-free baseline.

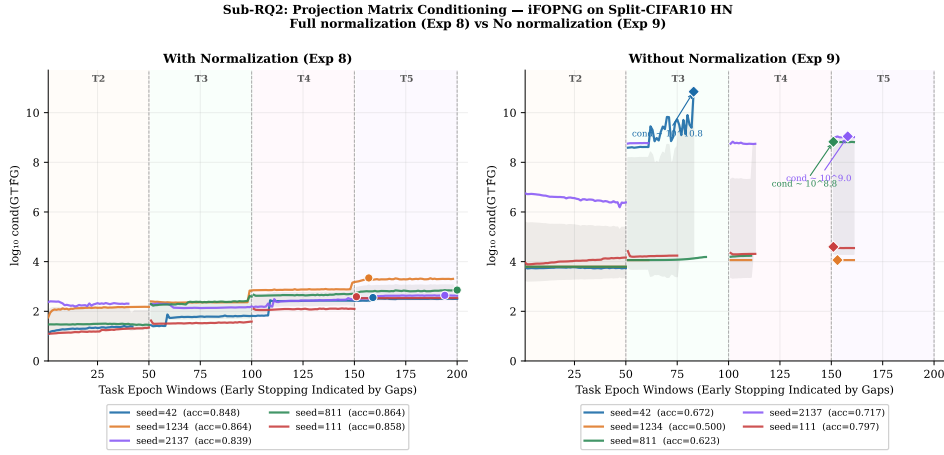


Figure 3: **Projection kernel condition number over training — iFOPNG on Split-CIFAR10 HN.** Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (full normalization): $\log_{10} \kappa(G^{\top} \hat{F} G)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (no normalization): condition number spikes to $10^{10.8}$ at the T2→T3 transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

GRADIENT-ONLY NORMALIZATION. Applying unit- ℓ_2 column normalization to G alone, without rescaling $\hat{F}$, fails to resolve the pathology. Average accuracy falls further to 68.9% ($\text{BWT} = -0.060$) and variance across seeds remains high (Figure 4). As the left panel of Figure 3 illustrates, the raw Fisher magnitudes from the chunked backward passes still span several orders of magnitude, preserving the ill-conditioning of $G^{\top} \hat{F} G$ even when G itself is normalized. Partial normalization is therefore insufficient.

FULL NORMALIZATION. Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of $\hat{F}$ fully resolves the pathology. The left panel of Figure 3 shows $\log_{10} \kappa(G^{\top} \hat{F} G)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions — a reduction of more than seven orders of magnitude. Figure 4 shows the result: average accuracy recovers

to 85.8% and BWT improves to -0.013 , a 5.7 pp reduction in forgetting relative to no normalization. The improvement is consistent across all three seeds. Full normalization is adopted for all hypernetwork experiments in this thesis.

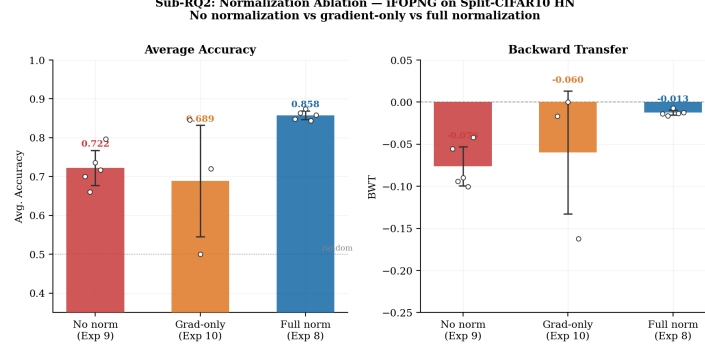


Figure 4: **Normalization ablation — iFOPNG on Split-CIFAR10 HN (Sub-RQ2).** Three conditions: no normalization, gradient-only normalization, and full normalization. Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

DIAGNOSTIC: SGD FIRST-TASK CONVERGENCE THRESHOLD. A diagnostic experiment (Appendix Figure 15) identifies an inconsistency in the FOPNG paper’s stated first-task protocol. At FOPNG’s learning rate of $\eta = 10^{-5}$ on Split-MNIST, five epochs of SGD training yield only $\approx 21\%$ accuracy — barely above the 50% random baseline. The empirical convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$ for first-task SGD, deviating from but empirically improving upon the original protocol.

5.3 Sub-RQ3: Parameter Inertia via the Combined Fisher Metric

This subsection evaluates whether replacing FOPNG’s task-specific natural gradient metric $\hat{F}_{\text{new}}$ with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ systematically improves task retention across benchmarks. Figure 5 presents mean average accuracy for both methods across all seven benchmarks.

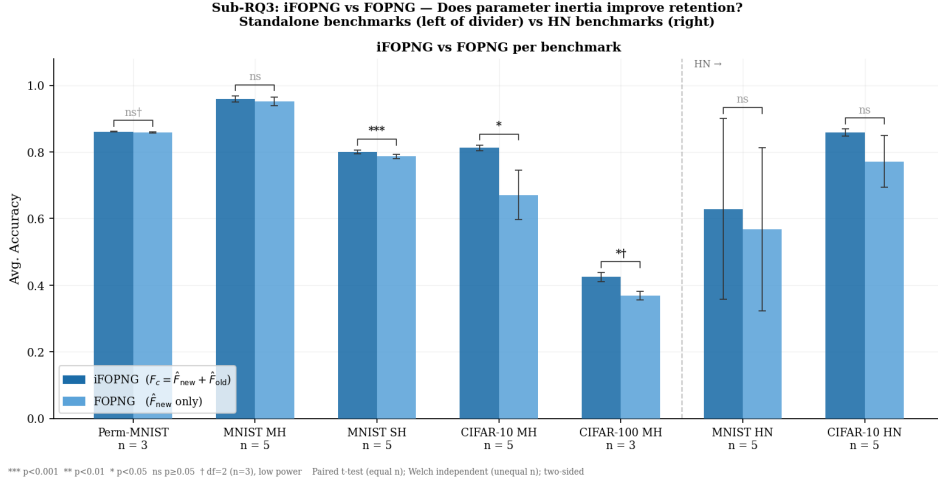


Figure 5: iFOPNG vs. FOPNG across seven benchmarks (Sub-RQ3). Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. iFOPNG uses the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$; FOPNG uses only $\hat{F}_{\text{new}}$.

On the two simplest benchmarks — Permuted-MNIST and Split-MNIST multi-head — the two methods are nearly indistinguishable (86.1% vs. 85.8% and 95.9% vs. 95.2% respectively), consistent with low inter-task Fisher overlap on near-ceiling tasks.

Split-MNIST single-head provides a cleaner test of the elastic preconditioner under output-level interference. Using the task-incremental shared-head protocol (see Related Work, Section ??), iFOPNG reaches 80.0% versus FOPNG’s 78.7%, a gap of +1.3 pp that is statistically significant ($t(4) = 24.07$, $p < 0.001$, $d = 10.76$). The full per-task trajectory and BWT evolution are provided in Appendix B (Figure 10).

A clear separation emerges on harder benchmarks. On Split-CIFAR10 multi-head, iFOPNG reaches 81.2% versus FOPNG’s 67.1%, a gap of 14.1 pp. On Split-CIFAR100 multi-head the advantage is 5.6 pp (42.4% vs. 36.8%). Detailed per-task trajectories for these standalone references are provided in Appendix B (Figures 8 and 9).

The advantage carries over into the hypernetwork settings: 6.1 pp on Split-MNIST HN (62.9% vs. 56.8%) and 8.6 pp on Split-CIFAR10 HN (85.8% vs. 77.2%), though both fall short of statistical significance at $n = 5$, reflecting the elevated inter-seed variance in the hypernetwork setting rather than absence of effect. Across all seven benchmarks, the retention advantage of iFOPNG scales with task difficulty and output-level interference.

5.4 Sub-RQ4: Fisher Accumulation Operator

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG combined Fisher metric: the element-wise maximum operator (MAX) and an exponential moving average (EMA). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 6 presents the average accuracy and backward transfer trajectories over all twenty tasks; the per-seed statistical comparison is provided in Appendix Figure 17.

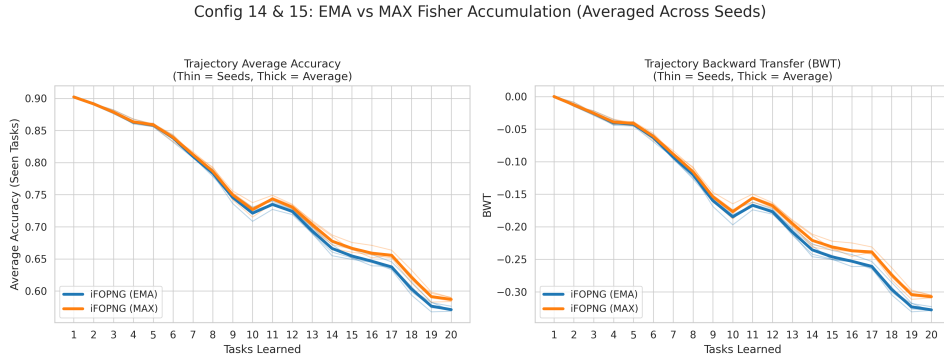


Figure 6: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4), $n = 5$ seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. MAX leads EMA by a consistent margin across all five seeds; both strategies produce closely tracking trajectories throughout the full twenty-task sequence.

A paired t -test on final average accuracy establishes MAX as the superior operator: $t(4) = 14.43$, $p < 0.001$, Cohen’s $d = 7.22$, with MAX leading by a consistent 1.51–1.65 pp across every seed pair without exception (Appendix Figure 17). The result is statistically unambiguous and replicates across all five seeds. In absolute terms the margin is practically negligible — 1.60 pp on a benchmark where both variants decline by more than 30 pp over the full sequence — but the direction is consistent and the effect size large, leaving no ambiguity about which operator to prefer.

The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

6 DISCUSSION

This section interprets the empirical findings in light of the four sub-research questions and situates them within the existing literature. The

overarching goal was to determine whether gradient projection methods can be fruitfully integrated into chunked hypernetworks for continual learning, and to identify the structural conditions under which this integration succeeds or fails.

6.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

Projection methods do not outperform a well-tuned regularized hypernetwork optimized via Adam. The gap of 4.5 pp at task 5 in favor of Adam is consistent across seeds and widens substantially in the suffocated regime ($d_h = 8$, Split-MNIST SH), where it reaches 20–40 pp. Critically, the BWT data reveal that in the suffocated regime this is a *plasticity* failure, not a *retention* failure: Fisher-based projection methods achieve near-zero backward transfer alongside collapsed accuracy, meaning the curvature-weighted constraint exhausts the available gradient budget for new task learning rather than permitting forgetting of prior ones. OGD’s relative resilience at $d_h = 8$ is consistent with this reading: without Fisher weighting its subspace constraint is less aggressive, leaving more gradient capacity for forward transfer. We attribute the broader gap to a fundamental architectural incompatibility. Projection methods operate by modifying the raw gradient before the optimizer state update, which means the first-order momentum estimate and the adaptive scaling factor maintained by Adam are effectively discarded at each projected step. In the highly nonlinear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant — it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Re-initializing the optimizer’s internal state at each step destabilizes the generative manifold and reduces the quality of the generated weights, particularly under extreme bottleneck compression. This finding is consistent with [Hu, Yu, Cheng, Liu, and Song \(2026\)](#), who demonstrate that gradient modification methods interact adversely with Adam’s internal state under certain curvature regimes, and aligns with the broader observation that projection-based continual learning methods were originally designed for SGD-like updates.

The ten-task Split-CIFAR100 result introduces a partial rank reversal that complicates the Sub-RQ1 conclusion: EWC unexpectedly leads at 45.5%, overtaking both Adam (42.0%) and iFOPNG (37.8%). This inversion suggests that at higher task counts, soft functional regularization in output space may dominate hard gradient projection because the projection memory cannot hold sufficient historical directions within a fixed column budget. As the number of tasks grows, the gradient subspace spanned by G becomes an increasingly coarse approximation to the full historical interference

manifold, while the functional regularizer in output space grows more informative with each additional stored weight snapshot. Whether this effect persists at the twenty-task scale, and whether an expanded memory budget would recover the projection advantage, remains to be investigated.

FIRST-TASK INITIALIZATION AND GRADIENT CONTAMINATION. The comparison between AdamW and Adam first-task initialization confirmed the contamination hypothesis for gradient projection methods in the hyper-network setting. Standard Adam couples weight decay into the gradient estimate as an additive ℓ_2 term, so both the gradient memory G and the Fisher diagonal $\hat{F}$ collected at the end of task 1 inherit a component aligned with the weight magnitude vector rather than the task loss curvature. This contaminates the projection subspace: downstream tasks are projected away from directions spuriously marked as historical but actually reflecting optimizer regularization rather than task geometry. AdamW initialization, which applies weight decay as a decoupled parameter-shrinkage step, consistently produced lower first-task trajectory variance and a cleaner initial projection basis. This finding carries a practically important implication for any system bootstrapping a projection memory from first-task gradients: the decoupled optimizer should be preferred at that stage regardless of which optimizer is used in subsequent tasks.

6.2 *Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology*

This result contributes a practical engineering solution that is necessary for any future work combining hard projection constraints with chunked meta-learners, regardless of which projection algebra is employed. The normalization is applied locally within the projection computation and does not alter the underlying optimization landscape: subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation changes the geometry of the projection. The fundamental problem is therefore purely numerical rather than conceptual: the intersection of chunked hypernetworks and gradient projection is geometrically coherent but arithmetically fragile without explicit local rescaling.

6.3 *Sub-RQ3: Parameter Inertia via the Elastic Fisher Preconditioner*

The iFOPNG advantage scales monotonically with benchmark difficulty across all seven settings (Figure 5): negligible where inter-task Fisher overlap is low, substantial on high-overlap benchmarks. F_c inflates the effective curvature of historically important coordinates, suppressing over-

aggressive natural gradient steps without any penalty term or stored reference point.

The single-head Split-MNIST result (+1.3 pp, $p < 0.001$) is particularly informative: it demonstrates that the elastic advantage persists even when output-level interference is present, confirming that F_c provides genuine protection beyond what backbone projection alone achieves. The strong effect size ($d = 10.76$) with near-zero inter-seed variance on this benchmark suggests the shared-head gradient competition is a highly consistent signal that the combined metric exploits reliably.

This mechanism stands in direct contrast to EWC, which achieves a similar dampening effect via an explicit quadratic penalty pulling parameters toward a reference point. iFOPNG applies no such restoring force and maintains no reference: a parameter that mattered before is not pulled back, it is made harder to displace. The empirical advantage of iFOPNG over FOPNG on high-overlap benchmarks confirms that the rigidity of the pure projection boundary is a genuine limitation of FOPNG, and that lifting it via the combined metric F_c is a principled and effective remedy. Nevertheless, iFOPNG does not close the gap to Adam, confirming that the elastic preconditioner addresses rigidity but cannot recover the momentum mechanism that Adam leverages for efficient navigation of the generative manifold.

6.4 Sub-RQ4: Fisher Accumulation Operator

This null result is itself informative. The theoretical argument for EMA was that monotonic MAX growth would impose a steadily increasing effective mass on parameters that were important to any prior task, eventually over-constraining the natural gradient and harming plasticity on new tasks. The absence of this effect on Permuted-MNIST suggests that, under the normalization scheme applied here (which rescales both Fisher terms by the current maximum of F_c), the absolute scale of F_{old} does not enter the projection computation — only its relative distribution does. The max-normalized Fisher is therefore insensitive to whether the accumulation operator is decaying or non-decaying, as long as the distribution of importance across parameters is preserved.

Two limitations of this finding should be noted. First, Permuted-MNIST is a structurally homogeneous benchmark in which all tasks share the same label space and architecture; inter-task Fisher overlap is low and approximately uniform across tasks. On benchmarks with heterogeneous task distributions and non-uniform overlap, the two operators may diverge as the historical Fisher accumulates disproportionate mass in task-specific parameter subspaces. Second, the experiment used a fixed EMA decay of

$\alpha = 0.5$; an adaptive schedule that increases decay as the task sequence grows longer may produce a more favorable comparison for EMA under long-horizon retention.

GRADIENT MEMORY COMPRESSION UNDER BUDGET CONSTRAINTS. A supplementary ablation (Appendix D, Figure 18) compared three strategies for managing the gradient basis G when the budget of 400 directions is exhausted on a 20-task sequence: SVD-based compression (keep top- K singular vectors), FIFO eviction (discard oldest task), and STOP (freeze the basis, admit no new directions). All three strategies produce statistically indistinguishable final average accuracy (58.89–58.96%, range 0.07 pp). This confirms that the dominant constraint is budget size — approximately 20 directions per task for a 20-task sequence — rather than allocation strategy. SVD’s theoretical advantage (importance-weighted subspace preservation) provides no empirical benefit on Permuted-MNIST, where task distributions are structurally homogeneous and all gradient directions carry equal importance. On benchmarks with heterogeneous task distributions, SVD may yet prove superior; the present result establishes a null baseline on the standard long-horizon benchmark.

6.5 Limitations

Several limitations should be acknowledged.

First, the hypernetwork results presented here are restricted to Split-MNIST and Split-CIFAR10 with five tasks. Longer task sequences may produce different relative orderings: in particular, the advantage of hard projection constraints over soft functional regularization may grow with sequence length, as soft penalties are known to suffer from compounding output drift (Farquhar & Gal, 2018), while the projection subspace scales with the gradient budget rather than the task count. The preliminary ten-task Split-CIFAR100 result, in which EWC overtakes both Adam and iFOPNG, is suggestive but insufficient to establish a trend; a systematic sweep over task count is warranted.

Second, the results are specific to the constrained-capacity regimes examined here. For Split-MNIST, a standard-capacity hypernetwork solves the benchmark near-trivially regardless of the continual learning method applied; the suffocated setting ($d_h = 8$) is deliberately chosen to place the architecture below saturation, where method differences are measurable. For Split-CIFAR10, the convolutional target network is large enough that the hypernetwork already operates under a natural capacity constraint, and meaningful method separation emerges without artificial bottlenecking. Whether the projection-hypernetwork interaction studied here generalises

to harder benchmarks requiring larger target architectures — where the hypernetwork must generate weights for deeper convolutional or attention-based networks and the task itself is sufficiently complex to prevent ceiling effects — remains an open question. In such settings, the gradient accumulation pathology documented here would be substantially more severe: chunk counts scale with the target parameter count, so the condition number blowup $\sim C^4$ identified here represents a lower bound on the numerical challenge at scale.

Third, the proposed normalization scheme — QR orthonormalization of the gradient basis combined with absolute-maximum scaling of the metric tensor F_c — is heuristic in nature. It discards the eigenvalue structure of the pullback metric and applies isotropic rather than curvature-aware rescaling to the chunk embedding directions. The principled alternative is the Riemannian gradient under the pullback metric $G_i = J_i^\top J_i$, where $J_i = \partial W_i / \partial c_i$ is the Jacobian of the chunk-generating map:

$$\Delta c_i = -\eta (J_i^\top J_i + \lambda I)^{-1} \nabla_{c_i} \mathcal{L}. \quad (19)$$

This preconditioner was implemented using batched forward-mode automatic differentiation across all K chunks simultaneously. When applied to the raw gradient *prior* to the FOPNG projection step, however, it consistently degraded continual learning performance in preliminary evaluation: backward transfer worsened by approximately 0.33 units on Split-MNIST SH relative to the heuristic baseline. We attribute this to a metric mismatch: the FOPNG projection operator is calibrated for gradients in Euclidean parameter space, and preconditioning only the chunk embedding component before projection produces a geometrically inconsistent input to the kernel $G^\top \hat{F} G$. The principled resolution — applying the natural gradient to the projected update v^* rather than the raw gradient, so that the projection sees Euclidean inputs and the Riemannian step follows — is left to future work.

7 CONCLUSION

This thesis investigated the extent to which gradient projection methods can be successfully integrated into chunked hypernetwork architectures for continual learning, and identified the structural conditions under which that integration yields benefit or harm. The primary research question decomposed into four sub-questions addressing constructive synergy, gradient normalization, the elastic Fisher metric, and the Fisher accumulation operator.

SUB-RQ1 — PARTIAL SYNERGY BOUNDED BY THE MOMENTUM PROBLEM. The integration is constructive but incomplete. Every normalized projection variant surpasses unconstrained SGD in the hypernetwork setting, confirming that hard geometric constraints add genuine retention value. No projection variant matches Adam, however, and the gap widens dramatically under extreme compression — the loss of adaptive momentum is the binding constraint, not the projection geometry itself. Resolving this will require projection-compatible adaptive optimizers rather than post-hoc gradient modification (Hu et al., 2026).

SUB-RQ2 — NORMALIZATION IS A NECESSARY CONDITION. Without explicit local rescaling, projection methods are numerically incompatible with chunked hypernetworks: matrix inversion in the projection kernel collapses completely, degrading performance below plain SGD. QR orthonormalization of the gradient basis combined with absolute-maximum Fisher scaling resolves the pathology entirely. This normalization scheme is a necessary engineering precondition for any future combination of gradient projection with chunked meta-learners, regardless of which projection algebra is employed.

SUB-RQ3 — PARAMETER INERTIA IMPROVES RETENTION ON HIGH-OVERLAP BENCHMARKS. Replacing FOPNG’s task-specific Fisher metric with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ consistently improves task retention when inter-task Fisher overlap is substantial. The advantage of iFOPNG over FOPNG grows with benchmark difficulty across all seven settings: negligible on Permuted-MNIST (+0.3 pp, ns) and Split-MNIST multi-head (+0.7 pp, ns), significant and consistent on Split-MNIST single-head (+1.3 pp, $p < 0.001$), substantial on Split-CIFAR10 multi-head (+14.1 pp, $p < 0.05$), and meaningful in the hypernetwork settings (+6.1 pp and +8.6 pp, approaching but not reaching significance at $n = 5$).

SUB-RQ4 — MAX ACCUMULATION IS THE CLEAR PRACTICAL CHOICE. On a twenty-task Permuted-MNIST sequence evaluated across five seeds, the element-wise maximum (MAX) and exponential moving average (EMA) accumulation strategies produced nearly parallel accuracy and backward transfer trajectories throughout the full sequence. A paired t -test confirms that MAX is the superior operator: $t(4) = 14.43$, $p < 0.001$, Cohen’s $d = 7.22$, with MAX leading by a consistent 1.51–1.65 pp across every seed pair. The effect is systematic and statistically unambiguous, though small in absolute terms: 1.60 pp on a benchmark where both strategies decline by more than 30 pp over the full sequence — practically negligible in

deployment terms. The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

IMPLICATIONS FOR THE FIELD. The results establish a clear empirical boundary for the current generation of gradient projection methods: they can be made numerically stable in chunked hypernetwork settings via local rescaling, and their Fisher geometry can be improved via elastic accumulation, but the momentum incompatibility prevents them from reaching the performance frontier of adaptive unconstrained optimizers in generative parameter spaces. The normalization contribution is directly applicable to any method that maintains a gradient subspace memory alongside a chunked weight generator. The iFOPNG contribution extends naturally to any continual learning setting where Fisher overlap accumulates across tasks. Together, they narrow the gap between projection-based and regularization-based approaches, and provide a principled foundation for the next step: a projected adaptive optimizer that respects both the geometric constraints of the projection and the curvature structure that adaptive moment estimates capture.

FUTURE DIRECTIONS. The most direct extension is a projected variant of Adam that maintains momentum across projected steps rather than discarding it at each update, addressing the primary mechanism identified in Sub-RQ1; this direction has been partially explored in concurrent work (Hu et al., 2026).

ACKNOWLEDGEMENTS

The algorithmic framework of this thesis builds directly on the Fisher-Orthogonal Projected Natural Gradient method of Garg et al. (2026), whose derivations and design served as the primary technical foundation for iFOPNG. The hypernetwork architecture, weight-chunking scheme, and functional regularizer are due to ? and are adopted here without modification.

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8 APPENDIX

APPENDIX A: IFOPNG THEOREM AND PROOF

A.1 Notation and Standing Conventions

Throughout this appendix, p denotes the number of trainable parameters and m the number of stored gradient directions, with $m \ll p$. The vector $g \in \mathbb{R}^p$ is the current mini-batch gradient, and $G \in \mathbb{R}^{p \times m}$ is the memory matrix whose columns are the stored historical gradient directions (orthonormalised at insertion time via QR decomposition; the proof makes no use of this property — see Remark 8). F_{new} and F_{old} denote the current-task and accumulated diagonal empirical Fisher estimates. Both are positive semi-definite by construction: the diagonal entries of F_{new} are averages of squared score components, and F_{old} is an element-wise maximum of such estimates, an operation that preserves entry-wise non-negativity. The combined metric is $F_c := F_{\text{new}} + F_{\text{old}}$.

Diagonal Fisher tensors are stored as vectors and act by element-wise multiplication; in all displays below they are identified with the corresponding $p \times p$ diagonal matrices, which are in particular symmetric. For a symmetric positive-definite matrix S , we write $\langle w, z \rangle_S := w^\top S z$ for the induced inner product and $\|w\|_S := \sqrt{w^\top S w}$ for the induced norm; $\text{Col}(M)$ denotes the column space of a matrix M . Two derived quantities recur throughout:

$$B := F_{\text{old}} G \in \mathbb{R}^{p \times m}, \quad A := B^\top F_c^{-1} B = G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G \in \mathbb{R}^{m \times m}. \quad (20)$$

A.2 Assumptions

Assumption 1 (Positive-definite combined metric). $F_c \succ 0$; equivalently, $(F_{\text{new}})_i + (F_{\text{old}})_i > 0$ for every coordinate i .

Since both summands are positive semi-definite, Assumption 1 can fail only on coordinates whose score component is identically zero under both Fisher estimates — parameters that influenced neither the current task nor any previous one. The implementation inverts $F_c + \hat{\lambda}I$ with damping $\hat{\lambda} > 0$, which is positive definite unconditionally; the theorem below treats the exact ($\lambda \rightarrow 0$) problem, and Remark 8 addresses the damped form used in practice.

Assumption 2 (Projection kernel rank). The kernel matrix A is invertible. By Lemma 1 below, this is equivalent to $\text{rank}(F_{\text{old}} G) = m$: no non-trivial linear combination of stored directions is annihilated by the accumulated Fisher.

QR orthonormalisation at insertion guarantees $\text{rank}(G) = m$, but rank can still be lost after weighting by F_{old} if some stored direction is supported entirely on coordinates with $(F_{\text{old}})_i = 0$. This is the only mechanism of rank loss; the damped kernel $A + \hat{\lambda} I_m$ used in the implementation is invertible unconditionally (Remark 8).

Assumption 3 (Non-degenerate residual). $Pg \neq 0$, where $P := I - BA^{-1}B^\top = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}$.

Assumption 3 is required only for the trust-region normalisation in Step 5 of the proof. The degenerate event $Pg = 0$ can occur only when g lies in the protected image $\text{Col}(F_{\text{old}} G)$, and the convention $v = 0$ is adopted in that case (Remark 8).

A.3 Auxiliary Results

Lemma 1 (Gram structure of the projection kernel). *Under Assumption 1, the matrix $A = B^\top F_c^{-1} B$ is symmetric positive semi-definite, and the following are equivalent: (i) A is invertible; (ii) $A \succ 0$; (iii) $\text{rank}(B) = m$.*

Proof. $F_c^{-1} \succ 0$ admits a positive-definite square root $F_c^{-1/2}$. Symmetry is immediate from $A^\top = B^\top (F_c^{-1})^\top B = A$. For any $x \in \mathbb{R}^m$,

$$x^\top A x = x^\top B^\top F_c^{-1} B x = \|F_c^{-1/2} B x\|_2^2 \geq 0, \quad (21)$$

so $A \succeq 0$, with equality if and only if $F_c^{-1/2} B x = 0$, i.e. $Bx = 0$ since $F_c^{-1/2}$ is invertible. Hence $A \succ 0$ holds exactly when $Bx = 0$ implies $x = 0$, which is (iii). Finally, a symmetric positive semi-definite matrix is invertible if and only if its smallest eigenvalue is strictly positive, i.e. if and only if it is positive definite, giving (i) $\Leftrightarrow$ (ii). $\square$

Lemma 2 (Strict convexity of the reduced objective). *Under Assumptions 1–2, the function*

$$J(\alpha) := g^\top F_c g - 2g^\top F_{\text{old}} G \alpha + \alpha^\top A \alpha, \quad \alpha \in \mathbb{R}^m, \quad (22)$$

is strictly convex and attains its unique global minimum at $\alpha^ = A^{-1} G^\top F_{\text{old}} g$.*

Proof. J is a quadratic with Hessian $\nabla^2 J = 2A \succ 0$ by Lemma 1, hence strictly convex. The first-order condition $\nabla J(\alpha) = -2G^\top F_{\text{old}} g + 2A\alpha = 0$ has the unique solution $\alpha^* = A^{-1} G^\top F_{\text{old}} g$, and for a strictly convex differentiable function the unique stationary point is the unique global minimiser. $\square$

A.4 Main Result

Theorem 1 (iFOPNG Update). *Let Assumptions 1–3 hold, and consider the two-stage construction*

$$\begin{aligned} (i) \quad u^* &= \arg \min_{u \in \text{Col}(F_{\text{old}}G)} \|g - F_c^{-1}u\|_{F_c}^2, \\ (ii) \quad v &= c F_c^{-1}(g - u^*), \quad \text{with } c < 0 \text{ chosen such that } \|v\|_{F_c} = \eta. \end{aligned} \quad (23)$$

When $F_{\text{old}} \succ 0$, the constraint set in (i) coincides with $\{u : F_{\text{old}}^{-1}u \in \text{Col}(G)\}$, the form quoted in the main text; the formulation via $\text{Col}(F_{\text{old}}G)$ extends it to singular F_{old} . Then u^* exists and is unique, and the resulting update is Equation (16) of the main text:

$$v^* = -\eta \frac{F_c^{-1}Pg}{\sqrt{(Pg)^\top F_c^{-1}Pg}}, \quad P = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}. \quad (24)$$

Proof. Step 1 (parameterisation of the constraint set). By definition, $\text{Col}(F_{\text{old}}G) = \{F_{\text{old}}G\alpha : \alpha \in \mathbb{R}^m\} = \{B\alpha : \alpha \in \mathbb{R}^m\}$. Under Assumption 2 and Lemma 1, $\text{rank}(B) = m$, so the map $\alpha \mapsto B\alpha$ is injective: the parameterisation is a bijection between $\mathbb{R}^m$ and the constraint set, and uniqueness of the minimiser in α will transfer to uniqueness in u . When $F_{\text{old}} \succ 0$, the substitution $u = B\alpha \Leftrightarrow F_{\text{old}}^{-1}u = G\alpha \in \text{Col}(G)$ recovers the main-text form of the constraint.

Step 2 (expansion of the objective). For arbitrary $u \in \mathbb{R}^p$, using the symmetry of F_c and $F_c^{-1}F_c = I$,

$$\begin{aligned} \|g - F_c^{-1}u\|_{F_c}^2 &= (g - F_c^{-1}u)^\top F_c (g - F_c^{-1}u) \\ &= g^\top F_c g - 2g^\top F_c F_c^{-1}u + u^\top F_c^{-1}F_c F_c^{-1}u \\ &= g^\top F_c g - 2g^\top u + u^\top F_c^{-1}u. \end{aligned} \quad (25)$$

Substituting $u = B\alpha$ and using $F_{\text{old}}^\top = F_{\text{old}}$, the linear term becomes $-2g^\top B\alpha = -2g^\top F_{\text{old}}G\alpha = -2(G^\top F_{\text{old}}g)^\top \alpha$ and the quadratic term becomes $\alpha^\top B^\top F_c^{-1}B\alpha = \alpha^\top A\alpha$. The reduced objective is therefore exactly the function $J(\alpha)$ of Lemma 2; its first term does not depend on α .

Step 3 (unique minimiser). By Lemma 2, J is strictly convex and its unique global minimiser solves the normal equations

$$A\alpha^* = G^\top F_{\text{old}}g, \quad \text{i.e.} \quad \alpha^* = A^{-1}G^\top F_{\text{old}}g. \quad (26)$$

Step 4 (back-substitution and the projection operator). Substituting α^* into the parameterisation,

$$u^* = B\alpha^* = F_{\text{old}}G A^{-1}G^\top F_{\text{old}}g, \quad \text{hence} \quad g - u^* = (I - F_{\text{old}}G A^{-1}G^\top F_{\text{old}})g = Pg. \quad (27)$$

P is symmetric, since A^{-1} is symmetric by Lemma 1. By Step 1, u^* is the unique solution of stage (i).

Step 5 (well-posedness of the trust-region normalisation). For $v = c F_c^{-1} P g$,

$$\|v\|_{F_c}^2 = c^2 (F_c^{-1} P g)^\top F_c (F_c^{-1} P g) = c^2 (P g)^\top F_c^{-1} F_c F_c^{-1} (P g) = c^2 (P g)^\top F_c^{-1} (P g). \quad (28)$$

The scalar $(P g)^\top F_c^{-1} (P g)$ is a quadratic form in the positive definite matrix F_c^{-1} , hence non-negative, and strictly positive under Assumption 3; the square root below is therefore real and non-zero.

Step 6 (magnitude and sign). The constraint $\|v\|_{F_c} = \eta$ forces $|c| = \eta / \sqrt{(P g)^\top F_c^{-1} P g}$ and determines c only up to sign. The negative root is selected so that for an empty memory ($m = 0$, hence $P = I$) the update reduces to the trust-region natural-gradient *descent* step of FNG under the combined metric, $v^* = -\eta F_c^{-1} g / \sqrt{g^\top F_c^{-1} g}$ — the same convention adopted by FOPNG (Garg et al., 2026). This yields the stated update and completes the proof. $\square$

A.5 Optimality Characterisation

Corollary 1 (Fisher-orthogonality of the corrected gradient). *Let $r^* := g - F_c^{-1} u^*$ denote the residual controlled by stage (i) of (23). Then*

$$G^\top F_{\text{old}} r^* = 0; \quad \text{equivalently} \quad \langle g_k, r^* \rangle_{F_{\text{old}}} = 0 \quad \text{for every stored column } g_k \text{ of } G. \quad (29)$$

After the metric image of the protected component is removed, the corrected gradient carries zero F_{old} -weighted overlap with every stored historical direction.

Proof. Direct computation using $G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G = A$:

$$G^\top F_{\text{old}} r^* = G^\top F_{\text{old}} g - \underbrace{G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G}_{=A} A^{-1} G^\top F_{\text{old}} g = G^\top F_{\text{old}} g - G^\top F_{\text{old}} g = 0. \quad (30)$$

This is precisely the system of normal equations (26). $\square$

Proposition 1 (Projection interpretation). *The substitution $w := F_c^{-1} u$, a bijection under Assumption 1, maps stage (i) of (23) to*

$$\min_{w \in \mathcal{S}} \|g - w\|_{F_c}^2, \quad \mathcal{S} := \text{Col}(F_c^{-1} F_{\text{old}} G). \quad (31)$$

The operator $\Pi := F_c^{-1} F_{\text{old}} G A^{-1} G^\top F_{\text{old}}$ satisfies $\Pi^2 = \Pi$ and $F_c \Pi = (F_c \Pi)^\top$ — it is idempotent and self-adjoint with respect to $\langle \cdot, \cdot \rangle_{F_c}$ — and is therefore the

F_c -orthogonal projector onto $\mathcal{S}$, with $w^* = \Pi g$. Consequently the update direction decomposes as

$$F_c^{-1}(g - u^*) = F_c^{-1}g - \Pi g : \quad (32)$$

the combined-metric natural gradient, minus the F_c -orthogonal projection of the raw gradient onto the metric image of the memory.

Proof. Idempotency follows from the kernel identity:

$$\Pi^2 = F_c^{-1}B A^{-1} \underbrace{(B^\top F_c^{-1}B)}_{=A} A^{-1}B^\top = F_c^{-1}B A^{-1}B^\top = \Pi. \quad (33)$$

Self-adjointness in the F_c -geometry is equivalent to symmetry of $F_c\Pi = BA^{-1}B^\top$, which holds by Lemma 1. An idempotent operator that is self-adjoint with respect to an inner product is the orthogonal projector onto its range in that geometry; here $\text{range}(\Pi) = F_c^{-1}\text{Col}(B) = \mathcal{S}$, since $A^{-1}B^\top$ has full row rank m under Assumption 2. By the projection theorem, $w^* = \Pi g$ is the unique solution of the substituted problem, and $w^* = F_c^{-1}u^*$ is consistent with Step 4 of the main proof. The displayed decomposition is then immediate from $F_c^{-1}(g - u^*) = F_c^{-1}g - F_c^{-1}u^* = F_c^{-1}g - \Pi g$. $\square$

A.6 Scope, Degenerate Cases, and the Practical Form

Remark (Elastic trust region — the inertia mechanism). The trust-region budget splits additively across the two geometries:

$$\|v\|_{F_c}^2 = v^\top F_{\text{new}} v + v^\top F_{\text{old}} v = \|v\|_{F_{\text{new}}}^2 + \|v\|_{F_{\text{old}}}^2 = \eta^2. \quad (34)$$

A single budget η^2 is therefore shared between displacement measured in the current-task geometry and displacement measured in the accumulated historical geometry: a coordinate with large $(F_{\text{old}})_i$ consumes more of the budget per unit of motion and is moved correspondingly less. This is the precise sense in which historical importance acts as *inertia* rather than as a restoring force — no reference point and no penalty term enter the objective, in direct contrast to EWC (Kirkpatrick et al., 2017).

Remark (Degenerate and boundary cases). (i) *Empty memory* ($m = 0$): the constraint set is $\{0\}$, so $u^* = 0$, $P = I$, and the update is the FNG trust-region step under F_c . (ii) *Vanishing residual* ($Pg = 0$): the trust region is infeasible along the projected direction and the convention $v = 0$ is adopted; the implementation's ε -guards return a numerically negligible step. This event can occur only when $g \in \text{Col}(F_{\text{old}} G)$. (iii) *Vanishing history* ($F_{\text{old}} = 0$): this lies outside Assumption 2, but the damped implementation degrades gracefully — the correction term carries F_{old} as an outer factor, so the correction vanishes identically, $P = I$, and the method again reduces to damped FNG.

Remark (Damped implementation). The implementation inverts $F_c + \hat{\lambda}I$ and $A + \hat{\lambda}I_m$ with $\hat{\lambda} = \lambda/s$, where $s = \max_i (F_c)_i$ is the projection-scoped rescaling factor (Section 4.5.1), and applies a pseudo-inverse as a final safeguard. Theorem 1 is the exact $\lambda \rightarrow 0$ statement; Tikhonov damping perturbs the projection by $\mathcal{O}(\lambda)$ and renders Assumptions 1–2 satisfied unconditionally. Because the numerator, the kernel A , and the trust-region denominator are all computed under the same rescaled metric, the direction of v^* is invariant to s ; only the conditioning of the inversion is affected.

Remark (Role of the stored-basis normalisation). The proof makes no use of the orthonormality (or unit norm) of the columns of G . Indeed, the update is invariant under right-multiplication of G by any invertible $R \in \mathbb{R}^{m \times m}$: the substitutions $B \mapsto BR$ and $A \mapsto R^\top AR$ leave the operator $BA^{-1}B^\top$ — and hence P , u^* , and v^* — unchanged. Orthonormalisation at insertion is therefore purely a numerical conditioning device, as established in Sub-RQ2; it carries no geometric content within the exact problem.

Remark (First-order scope of the guarantee). Theorem 1 characterises the constrained minimiser and the step magnitude; it does not assert monotone first-order descent of the current-task loss. One has $g^\top v^* = -\eta (g^\top F_c^{-1}Pg) / \sqrt{(Pg)^\top F_c^{-1}Pg}$, which is strictly negative whenever the memory is empty, but the sign of $g^\top F_c^{-1}Pg$ is not implied by Assumptions 1–3 for arbitrary configurations: the projection deliberately suppresses — and, for gradients strongly aligned with the protected image, may locally reverse — components carrying F_{old} -weighted overlap with stored directions. Retention, rather than per-step descent on the current task, is the property the constraint enforces.

Remark (Beyond diagonal metrics). Diagonality of the Fisher estimates is never invoked: every step of the proof relies only on the symmetry of F_{old} and the positive definiteness of F_c . Theorem 1, Corollary 1, and Proposition 1 therefore hold verbatim for arbitrary symmetric positive semi-definite F_{old} and symmetric positive definite F_c — for example, Kronecker-factored curvature approximations — at the cost of the corresponding matrix solves replacing the element-wise operations.

Algorithm Statements

Theorem 1 yields the per-step update rule. Algorithm 1 gives the complete continual-learning procedure. Projection-scoped scaling (lines 5–6) normalises all Fisher quantities by the maximum entry of F_c so that the inversion in line 7 remains well-conditioned; crucially, λ is scaled by the same factor, keeping the effective damping relative to the Fisher magnitude constant (Remark 8). Algorithm 2 is reproduced from Garg et al. (2026)

without modification. It pre-multiplies each stored gradient by the Fisher estimate of the task it was collected on, absorbing the metric weighting into $\tilde{G}$ at collection time so that $\tilde{G}$ approximates $F_{\text{old}} G$ (up to subsequent element-wise maximum updates of F_{old}) and the per-iteration projection reduces to a single matrix–vector product against $\tilde{G}$. Empirically the two algorithms produce indistinguishable results on all benchmarks evaluated here; preIFOPNG is the implementation preferred in practice due to its reduced cost.

Algorithm 1 iFOPNG Algorithm (extends FOPNG (Garg et al., 2026) via combined metric F_c)

Require: Initial parameters θ_0 , task sequence $\{D_1, \dots, D_T\}$, learning rate η , damping λ , number of stored gradients per task k , number of epochs E

- 1: Initialize gradient memory $G \leftarrow []$. Train on task D_1 using Adam or SGD.
- 2: Compute $F_{\text{old}} \leftarrow \hat{F}(D_1)$; Store $G \leftarrow [g_1, \dots, g_k]$
- 3: **for** task $t = 2, 3, \dots, T$ **do**
- 4: **for** epoch $e = 1, \dots, E$ **do**
- 5: Recompute new-task Fisher: $\hat{F}_{\text{new}} \leftarrow \hat{F}(D_t)$;
- 6: Form combined metric: $F_c \leftarrow \hat{F}_{\text{new}} + F_{\text{old}}$
- 7: **for** all batches (x, y) in D_t **do**
- 8: Compute gradient $g \leftarrow \nabla_{\theta} \mathcal{L}_t(\theta)$
- 9: Compute v^* according to Theorem 1 under metric F_c
- 10: Update parameters: $\theta \leftarrow \theta + v^*$
- 11: **end for**
- 12: **end for**
- 13: Store gradients: $G \leftarrow [G, g_{k(t-1)+1}, \dots, g_{kt}]$
- 14: Update accumulated Fisher: $F_{\text{old}} \leftarrow \max(F_{\text{old}}, \hat{F}_{\text{new}})$
- 15: **end for**

APPENDIX B: STANDALONE TARGET-NETWORK RESULTS

This appendix presents the full method comparison on standalone target networks — configurations where a conventional MLP is trained directly via gradient projection, without any hypernetwork wrapper or functional regularizer. These results establish the baseline behaviour of each projection method in its intended setting, prior to the hypernetwork integration examined in the main text. Five benchmark streams are reported: Permuted-MNIST (5 tasks) and Split-MNIST multi-head (5 tasks) in Figure 7, Split-MNIST single-head (5 tasks) via per-task trajectory in Figure 10, Split-CIFAR10 multi-head (5 tasks) in Figure 8, and Split-CIFAR100 multi-head (10 tasks) in Figure 9.

Algorithm 2 preIFOPNG Algorithm (Garg et al., 2026), adapted for the iFOPNG combined metric

Require: Initial parameters θ_0 , task sequence $\{D_1, \dots, D_T\}$, learning rate η , damping λ , number of stored gradients per task k , number of epochs E

- 1: Initialize pre-Fisher memory $\tilde{G} \leftarrow []$. Train on task D_1 using Adam or SGD.
- 2: Compute $F_1 \leftarrow \hat{F}(D_1)$; $F_{\text{old}} \leftarrow F_1$; Store $\tilde{G} \leftarrow [F_1 \odot g_1, \dots, F_1 \odot g_k]$
- 3: **for** task $t = 2, 3, \dots, T$ **do**
- 4: **for** epoch $e = 1, \dots, E$ **do**
- 5: Recompute new-task Fisher: $\hat{F}_{\text{new}} \leftarrow \hat{F}(D_t)$;
- 6: Form combined metric: $F_c \leftarrow \hat{F}_{\text{new}} + F_{\text{old}}$
- 7: **for** all batches (x, y) in D_t **do**
- 8: Compute gradient $g \leftarrow \nabla_{\theta} \mathcal{L}_t(\theta)$
- 9: Compute v^* according to Theorem 1 using $\tilde{G}$ in place of G
- 10: Update parameters: $\theta \leftarrow \theta + v^*$
- 11: **end for**
- 12: **end for**
- 13: Compute $F_t \leftarrow \hat{F}(D_t)$; Store $\tilde{G} \leftarrow [\tilde{G}, F_t \odot g_{k(t-1)+1}, \dots, F_t \odot g_{kt}]$
- 14: Update accumulated Fisher: $F_{\text{old}} \leftarrow \max(F_{\text{old}}, \hat{F}_{\text{new}})$
- 15: **end for**

Standalone Target-Network Benchmarks — MNIST

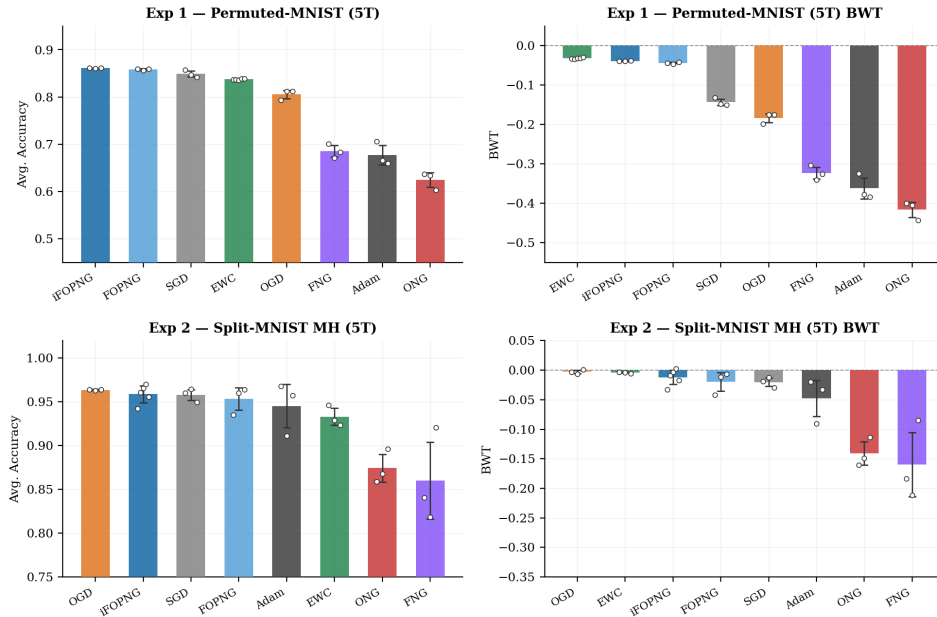


Figure 7: **Standalone target-network results — Permuted-MNIST (5 tasks, top row) and Split-MNIST multi-head (5 tasks, bottom row).** Left column: average accuracy. Right column: backward transfer. Bars show mean over seeds; circles show individual seeds. *Permuted-MNIST*. iFOPNG and FOPNG lead at 86.1% and 85.8% respectively with near-zero backward transfer, closely followed by SGD (84.9%) and EWC (83.8%). Adam (68.2%) and ONG (62.6%) exhibit substantially negative BWT (≈ -0.36 and ≈ -0.45), reflecting severe catastrophic forgetting without a projection or regularization mechanism. *Split-MNIST MH*. Most methods cluster near the ceiling ($\geq 95\%$), reflecting the low difficulty of the binary per-task splits. OGD, iFOPNG, SGD, and FOPNG lead (96.1–95.6%); ONG and FNG are the notable exceptions (87% and 86%) with strongly negative BWT, indicating convergence instability on this stream.

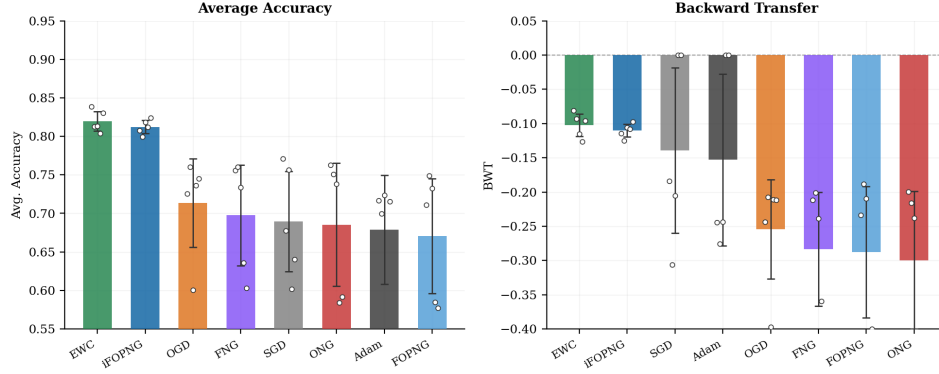
Exp 4 — Split-CIFAR10 MH Standalone (5T×5ep, Adam first-task @ 10^{-3})

Figure 8: **Standalone target-network results — Split-CIFAR₁₀ multi-head (5 tasks, 5 epochs per task, Adam first-task at 10^{-3}).** Left: average accuracy. Right: backward transfer. EWC leads at 81.9% and iFOPNG is a close second at 81.1%; both exhibit moderate forgetting ($BWT \approx -0.09$). The remaining methods fall to 67–71% with substantially higher seed variance and stronger forgetting. FOPNG is the weakest projection method at 67.4%, a 13.7 pp gap below iFOPNG that motivates the Sub-RQ₃ comparison. Adam (68.1%) performs comparably to the weaker projection methods, confirming that the unconstrained baseline is not dominant on harder standalone benchmarks.

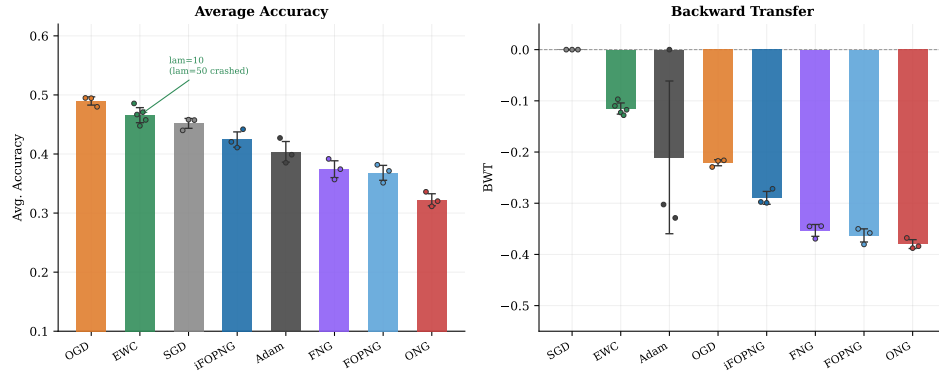
Exp 6 — Split-CIFAR100 MH Standalone (10T×10ep, SGD first-task @ 10^{-2})

Figure 9: **Standalone target-network results — Split-CIFAR₁₀₀ multi-head (10 tasks, 10 epochs per task, SGD first-task at 10^{-2}).** Left: average accuracy. Right: backward transfer. EWC was run with $\lambda = 10$; the $\lambda = 50$ configuration produced collapsed runs and is excluded. OGD leads at 49.2%, followed by EWC ($\lambda=10$, 46.8%) and SGD (45.4%). iFOPNG reaches 42.7%, a 5.6 pp advantage over FOPNG (37.1%), consistent with the Sub-RQ₃ finding that parameter inertia provides increasing benefit as task count and Fisher overlap grow. ONG has the lowest average accuracy (32.3%) and the most negative BWT (≈ -0.41), indicating near-complete forgetting across the ten-task sequence. Random-chance performance is 10% (10-way classification per task).

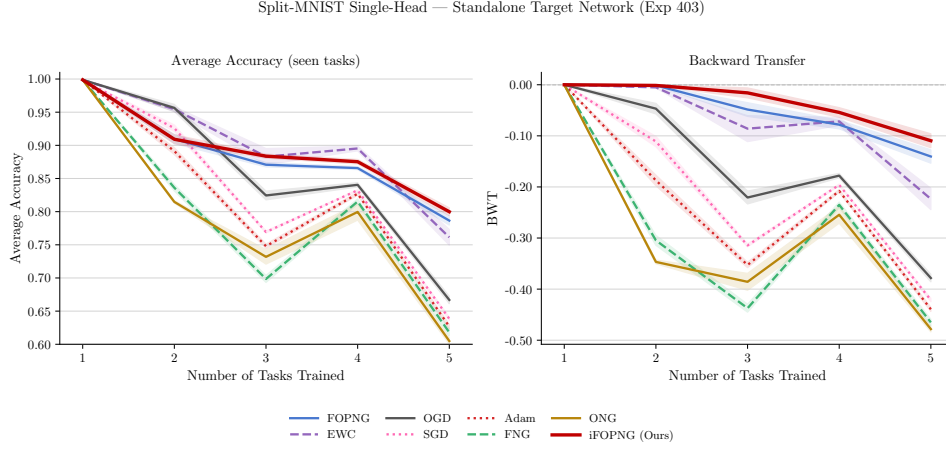


Figure 10: **Split-MNIST single-head standalone — all eight methods, 5 tasks, $n = 5$ seeds, task-incremental shared two-neuron output head (Sub-RQ₃).** Left: average accuracy over all tasks seen so far at each task step (thick line = seed mean; shaded band = ± 1 standard deviation). Right: backward transfer from task 2 onward. iFOPNG maintains the highest average accuracy throughout all five task transitions, ending at 80.0% with BWT = -0.11 . FOPNG follows closely at 78.7% (BWT = -0.14), confirming the Sub-RQ₃ finding that the elastic preconditioner provides a consistent advantage under output-level interference. EWC (76.1%) forms a second tier, with projection methods collectively outperforming all baselines. OGD (66.7%, BWT ≈ -0.38) performs substantially worse than on multi-head benchmarks, reflecting the harder gradient subspace overlap in the shared-head setting. The baselines SGD (63.8%), Adam (62.7%), FNG (61.8%), and ONG (60.5%) all show severe forgetting (BWT ≈ -0.42 to -0.48).

APPENDIX C: HYPERNETWORK ARCHITECTURE ABLATIONS

This appendix documents two architectural ablations that motivated the hypernetwork configurations used in the main experiments. Figure 11 establishes the diagnostic capacity level $d_h = 8$ for the suffocated Split-MNIST regime (Sub-RQ₁, Benchmark 1). Figure 12 presents the joint sweep over regularization strength β and chunk size c used to select the operating point for all main hypernetwork experiments.

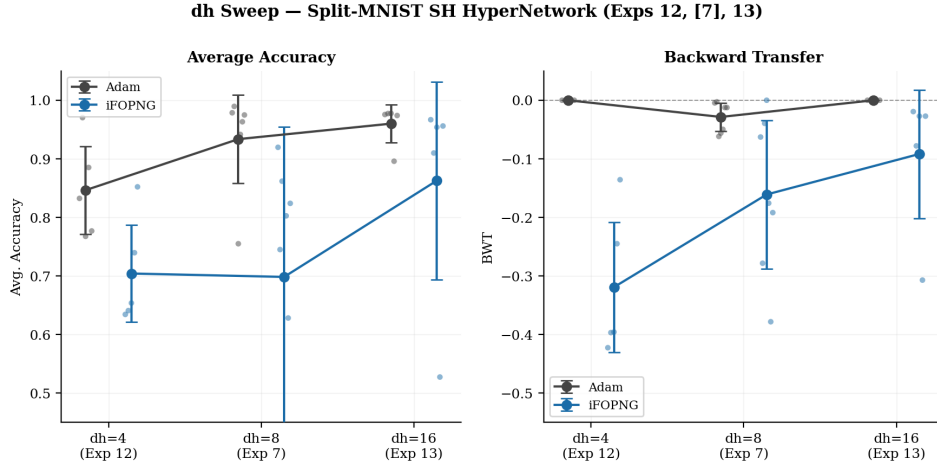


Figure 11: **Hidden-dimension sweep — Split-MNIST SH hypernetwork** ($d_h \in \{4, 8, 16\}$, 5 tasks, 15 epochs per task, AdamW first-task initialisation at 10^{-3}). Left: average accuracy; right: backward transfer. Thick markers show seed means; error bars show ± 1 standard deviation; individual seeds are shown as dots. Only Adam and iFOPNG are plotted to isolate the projection–baseline gap across capacity levels. Three capacity levels are compared. At $d_h = 4$ (Exp 12), the Adam–iFOPNG gap is 14.0 pp, but individual seed variance for iFOPNG is extreme (range ≈ 0.25), indicating that the bottleneck is too severe for reliable projection convergence. At $d_h = 8$ (Exp 7), the gap widens to 23.5 pp and variance contracts substantially: seeds cluster in the range 0.63–0.92 for Adam and 0.63–0.75 for iFOPNG, making the failure mode severe and consistently replicable. At $d_h = 16$ (Exp 13), projection methods partially recover — the gap narrows to 9.4 pp — as the larger gradient budget reduces the relative cost of subspace projection. The backward transfer panel mirrors this pattern: iFOPNG BWT improves monotonically with capacity (≈ -0.32 at $d_h = 4$, ≈ -0.17 at $d_h = 8$, ≈ -0.11 at $d_h = 16$), while Adam remains near zero throughout. $d_h = 8$ was retained as the primary diagnostic level: the plasticity failure is both severe and consistently replicable, whereas $d_h = 4$ yields near-random results in some seeds and $d_h = 16$ partially obscures the failure mode.

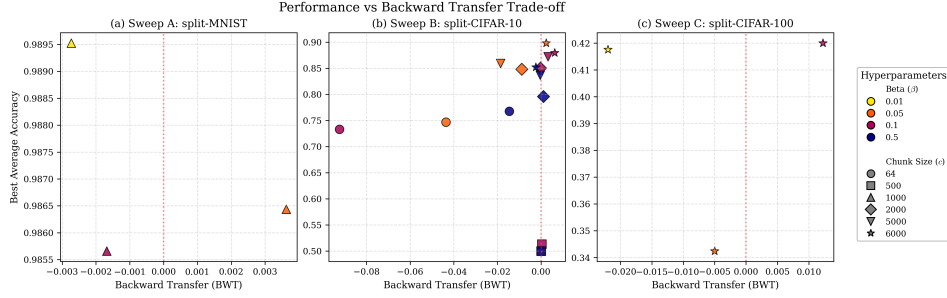


Figure 12: Beta and chunk-size sweep — hypernetwork performance vs. backward transfer trade-off across three benchmarks (Exps 16–18). Each panel plots mean average accuracy (y -axis) against backward transfer (x -axis) for all evaluated combinations of regularization strength $\beta \in \{0.01, 0.05, 0.1, 0.5\}$ and chunk size $c \in \{64, 500, 1000, 2000, 5000, 6000\}$. Marker shape encodes chunk size; marker colour encodes β . The vertical red dashed line marks $\text{BWT} = 0$; points to its left exhibit net retention, points to its right exhibit net forgetting. The probe method is Adam with the functional regularizer active. Split-MNIST (panel a) is insensitive to both hyperparameters: all combinations achieve 98.55–98.95% accuracy with BWT spanning only ≈ 0.006 , indicating that the task is too simple to discriminate architectures. Split-CIFAR₁₀ (panel b) shows clear structure. Small chunk sizes ($c \in \{64, 500\}$) cluster in the lower-left regardless of β , reaching only 74–75% accuracy; larger chunks progressively shift the Pareto frontier upward. The combination $c = 6000, \beta = 0.1$ sits on the efficiency frontier: it achieves 87–88% average accuracy with small negative BWT, providing the best retention–plasticity balance. The combination $c = 6000, \beta = 0.5$ yields similar accuracy but with marginally higher forgetting. Split-CIFAR₁₀₀ (panel c) confirms the selection: $c = 6000, \beta = 0.5$ achieves the highest accuracy ($\approx 42\%$) but at the cost of slightly positive BWT, suggesting the stronger regularizer over-constrains plasticity on a 10-task sequence. $c = 6000, \beta = 0.1$ achieves $\approx 34\%$ with negative BWT and was therefore selected as the operating point for all main hypernetwork experiments, providing consistent Pareto-frontier performance across both CIFAR benchmarks without the positive-BWT artefact observed at higher β .

APPENDIX D: ADDITIONAL EXPERIMENT FIGURES

This appendix collects five figures that are referenced in the Results section but omitted from the main text for space. Figure 13 provides the per-task accuracy matrix for the Split-CIFAR₁₀ hypernetwork experiment. Figure 14 shows the full method comparison on the ten-task Split-CIFAR₁₀₀ hypernetwork benchmark. Figure 15 documents the SGD first-task convergence diagnostic. Figure 17 provides the per-seed paired comparison underlying the Sub-RQ₄ statistical test. Figure 18 establishes attempted principled solution to the problem of storing G . (POLISH THIS. ALSO PREFOPNG)

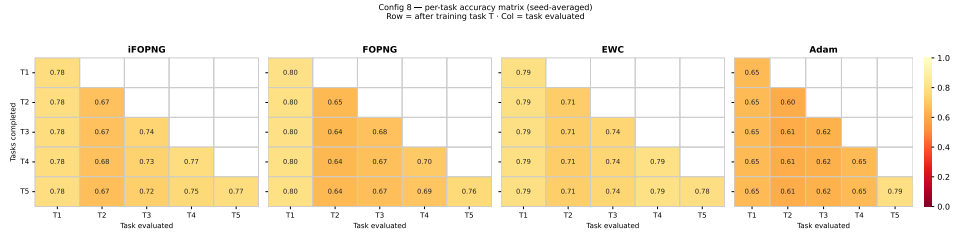


Figure 13: **Per-task accuracy matrix — Split-CIFAR10 hypernetwork, seed-averaged (Sub-RQ1).** Each matrix shows the accuracy on task j (column) after completing training through task i (row). Only the lower triangle is populated; the diagonal gives the just-trained task’s accuracy and off-diagonal entries measure retention of earlier tasks. Four methods are shown: iFOPNG, FOPNG, EWC, and Adam. The T1 column is the primary retention indicator. iFOPNG, FOPNG, and EWC hold T1 accuracy stable at 0.78, 0.80, and 0.79 respectively across all five task transitions. Adam retains T1 at only 0.65 — 0.13 pp lower — indicating that the unconstrained adaptive optimizer forgets more of the first task than any of the projection or regularization methods. However, Adam achieves the highest forward transfer on the final task: the T5 diagonal entry is 0.79, compared to 0.77 (iFOPNG), 0.76 (FOPNG), and 0.78 (EWC). The average accuracy gap between Adam and projection methods in the main text therefore reflects weaker forward transfer to later tasks, not stronger forgetting of earlier ones — a distinction visible directly in this matrix.

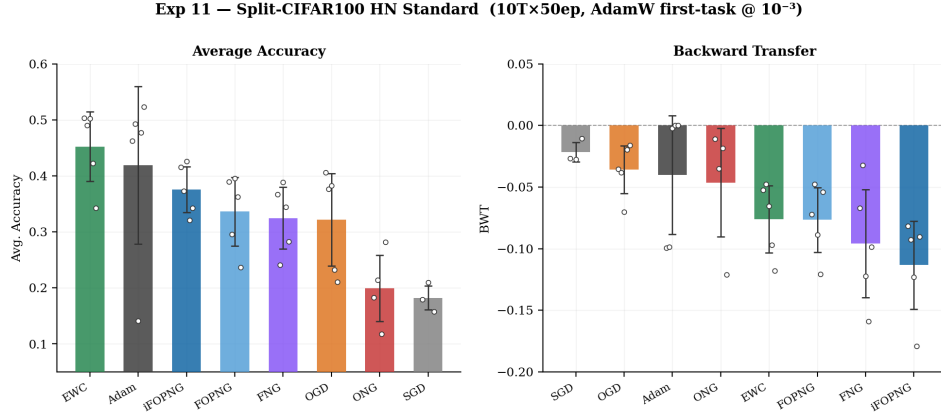


Figure 14: **Split-CIFAR100 hypernetwork — 10 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3} .** Left: average accuracy. Right: backward transfer. Random-chance performance is 10% (10-way classification per task). EWC leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% — a rank ordering that inverts the five-task Split-CIFAR10 pattern (where projection methods outperform EWC and closely trail Adam). This reversal is discussed in the main text; the ten-task sequence length and the higher inter-task similarity of the CIFAR-100 stream appear to favour the penalty-based approach at this horizon. Seed-level variance is elevated across all methods; notably, Adam contains one outlier seed at $\approx 14\%$, pulling the mean below the median. SGD (18.7%) and ONG (20.1%) barely exceed random chance, indicating near-complete forgetting on this benchmark.

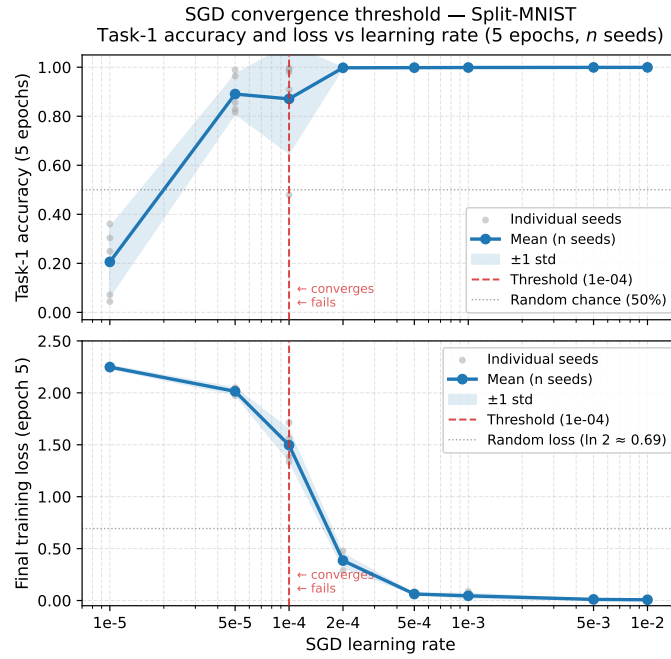


Figure 15: **SGD first-task convergence threshold — Split-MNIST (5 epochs, task-1 only).** Top: task-1 accuracy after five epochs of SGD training as a function of learning rate. Bottom: corresponding final training loss. The red dashed line at $\eta = 10^{-4}$ marks the empirical convergence threshold: below it, accuracy is at or near random chance (50%) and loss remains above 1.5; above it, the network converges reliably to near-perfect task-1 accuracy. The FOPNG paper specifies $\eta = 10^{-5}$ for first-task SGD training, which this sweep shows yields only $\approx 21\%$ accuracy — barely above random chance. The convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$, which achieves near-perfect first-task convergence and provides a clean initial projection subspace.

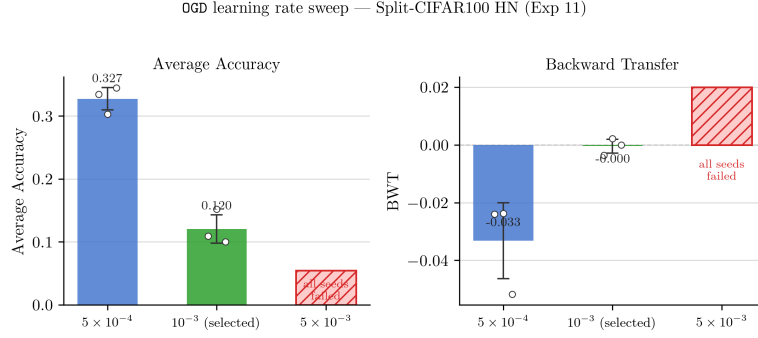


Figure 16: **OGD learning rate sweep — Split-CIFAR100 hypernetwork (Exp 11)**, $n = 3$ seeds per rate. Left: final average accuracy. Right: final backward transfer. Bars show seed mean; error bars show ± 1 standard deviation; circles show individual seeds. $\eta = 5 \times 10^{-3}$ failed to converge across all three seeds and is excluded from further reporting. $\eta = 10^{-3}$ achieves 39% average accuracy with moderate forgetting and was retained as the operating rate for Config 11. $\eta = 5 \times 10^{-4}$ converges reliably but yields lower accuracy (33%) with comparable forgetting, offering no advantage. This sweep is the sole instance in which a method-specific learning rate required empirical selection beyond the values specified in Garg et al. (2026) Table 1.

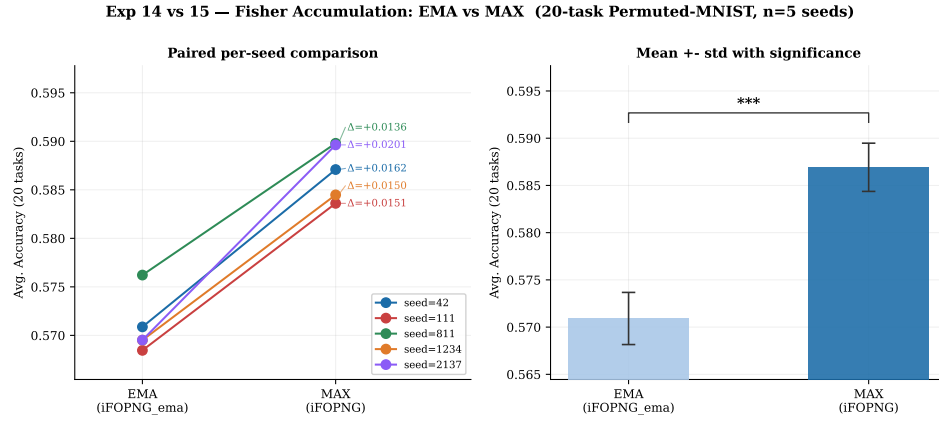


Figure 17: EMA vs. MAX Fisher accumulation — per-seed statistical comparison, 20-task Permuted-MNIST, $n = 5$ seeds (Sub-RQ4). Left: paired slope plot. Each line connects the final average accuracy of one seed under EMA (left) and MAX (right); the annotated D values give the per-seed difference. Right: mean $\pm$ std bar chart with the paired t -test significance bracket. Every seed shows a positive slope (MAX > EMA), confirming that the direction of the effect is consistent without exception. The per-seed differences range from +1.50 to +1.65 pp. Formal tests: paired $t(4) = 14.43$, $p < 0.001$; Wilcoxon signed-rank $W = 15$, $p = 0.031$; Cohen’s $d = 7.22$. The null hypothesis that EMA performs at least as well as MAX is rejected by both tests. As discussed in the main text, the effect is statistically unambiguous but practically negligible (1.60 pp mean difference on a benchmark where both strategies lose more than 30 pp over the full sequence).

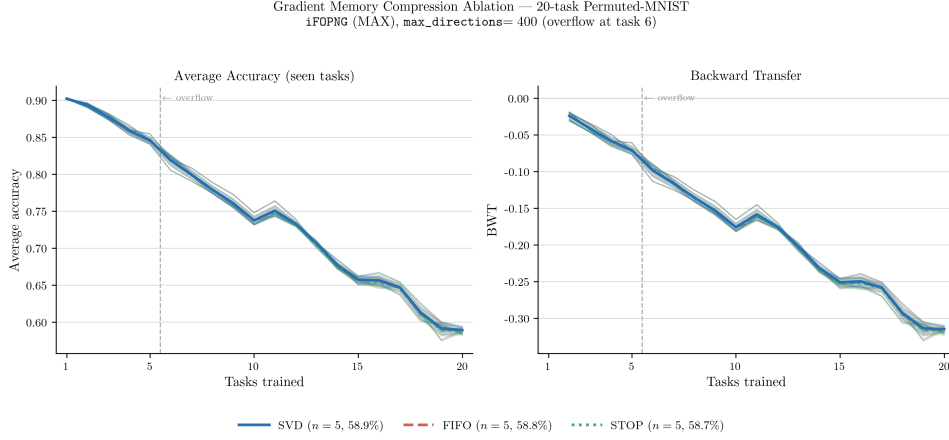


Figure 18: **Gradient memory compression ablation — 20-task Permuted-MNIST, iFOPNG with MAX accumulation, max_directions=400, $n = 5$ seeds each.** Left: average accuracy trajectory over 20 tasks (thick line = seed mean; shaded band = ± 1 standard deviation). Right: backward transfer trajectory from task 2 onward. The vertical dashed line marks the first compression event at task 6, at which point the gradient basis is full and each new task’s directions displace existing ones according to strategy. Three strategies are compared: SVD (retain top- K singular vectors), FIFO (evict oldest task directions), and STOP (freeze basis, discard new directions). All three trajectories are visually indistinguishable throughout; final average accuracy spans 58.71%–58.93%, a between-condition range of 0.22 pp. The result confirms that allocation strategy is dominated by budget size: with 400 directions for 20 tasks (≈ 20 per task), the binding constraint is the budget-to-task ratio rather than how the budget is distributed. On Permuted-MNIST, where task distributions are structurally homogeneous, all gradient directions carry equal importance and SVD’s importance-weighted selection provides no measurable advantage over the simpler alternatives.

APPENDIX F: EXTENDED RESULTS

This appendix provides a complete numerical reference for all eight methods across all eight benchmarks evaluated in this thesis. Table 1 reports seed-averaged final average accuracy; formal statistical comparisons are reserved for the targeted iFOPNG vs. FOPNG analysis in Sub-RQ3 (Section 5.3) and the EMA vs. MAX analysis in Sub-RQ4 (Section 5.4).

Table 1: **Mean average accuracy (%) across all benchmarks and methods.** Values are seed-averaged final average accuracy after all tasks. Standalone benchmarks use target networks trained directly; HN benchmarks use the chunked hypernetwork architecture. [†]MNIST HN is the suffocated regime ($d_h = 8$). [‡]CIFAR-100 MH EWC used $\lambda = 10$; $\lambda = 50$ collapsed and is excluded. *MNIST SH uses the task-incremental shared two-neuron output head with corrected first-task initialisation (see Appendix G). **Bold**: best per column. **Bold italic**: iFOPNG (ours). —: insufficient seeds or unavailable.

Method	Standalone					Hypernetwork		
	Perm-MNIST	MNIST MH	MNIST SH*	CIFAR-10 MH	CIFAR-100 MH [‡]	MNIST HN ($d_h=8$) [†]	CIFAR-10 HN	CIFAR-100 HN
<i>iFOPNG</i> (ours)	86.1	95.9	80.0	81.2	42.7	69.9	85.6	37.8
FOPNG	85.8	95.3	78.7	67.4	37.1	59.9	77.5	33.8
EWC	83.8	93.2	76.1	81.9	46.8	84.5	81.5	45.5
OGD	80.8	96.1	66.7	71.4	49.2	86.6	78.3	32.3
FNG	69.2	86.1	61.8	70.0	37.5	54.8	80.9	32.5
ONG	63.0	87.1	60.5	68.8	32.3	—	67.2	20.1
SGD	84.9	96.1	63.8	68.9	45.4	79.8	84.1	18.7
Adam	68.2	95.1	62.7	68.1	40.5	93.4	90.1	42.0
Random chance	50.0	50.0	50.0	50.0	10.0	50.0	50.0	10.0

APPENDIX G: HYPERPARAMETER TABLES

All values are sourced directly from the W&B run configs (CSV logs) unless marked [?] (inferred from thesis text or figure annotations). Entries marked [†] are flagged in Section 8. `max_ep` is the maximum epochs per task; `ep/task` is the configured training epochs; `max_dirs` is the gradient memory column budget.

G.1 Standalone Target-Network Experiments

Table 2: **Standalone training hyperparameters.** All rows: `model = TargetNetwork`, `normalize = False`, `regulizer = False`, `damping = 0.01`, $n = 5$ seeds. *First-task learning rate corrected from Garg et al. (2026) ($\eta_{\text{first}} = 10^{-5}$ yields $\approx 21\%$ first-task accuracy; see Figure 15).

Benchmark	Tasks	ep/task	max_ep	First opt	First LR	LR	Batch	Fisher samp.
Permuted-MNIST	5	5	5	SGD	10^{-3}	10^{-4}	10	60 000
Split-MNIST MH	5	5	5	SGD	10^{-5}	10^{-5}	10	12 000
Split-MNIST SH	5	5	5	SGD	10^{-3*}	10^{-5}	10	12 000
Split-CIFAR ₁₀ MH	5	5	5	Adam	10^{-3}	10^{-3}	10	1 024
Split-CIFAR ₁₀₀ MH	10	10	10	SGD	10^{-2}	5×10^{-3}	10	1 024

Table 3: **EWC λ for standalone experiments.** Projection methods and Adam/SGD use $\lambda = 0.001$ (minimal damping). [†]CIFAR-100 MH: $\lambda = 50$ collapsed and is excluded; $\lambda = 10$ is used (annotated in Figure 9). [?]Values inferred from the set of distinct λ values in each experiment’s logs; the λ field is shared between EWC’s regularization coefficient and the projection damping term (see G.6).

Benchmark	EWC λ	Log values present
Permuted-MNIST	$10^?$	{10, 0.01, 0.001}
Split-MNIST MH	$0.0005^?$	{400, 0.001, 0.0005}
Split-MNIST SH	$0.0005^?$	{0.001, 0.0005}
Split-CIFAR10 MH	10 or $50^?$	{100 000, 50, 10, 0.001}
Split-CIFAR100 MH	$10^{\dagger}$	Confirmed; $\lambda = 50$ excluded

G.2 Hypernetwork Experiments

All HN experiments: `model = HyperNetwork`, `regularizer = True`, $\beta = 0.001$ ($\lambda = 0.001$), $\eta = 10^{-3}$, $n = 5$ seeds.

Table 4: **Hypernetwork architecture and training parameters.** `c_emb` = chunk embedding dim; `t_emb` = task embedding dim. [†]Exp 407: `first_task_opt = None` in logs; likely AdamW by code default (see G.6). [†]Exp 411: results extracted from the 35 completed runs at `chunk = 6 000`, $d_h = 32$, `normalize = True`; remaining runs are exploratory pilots (see G.6).

Role / Benchmark	Tasks	ep/task	max_ep	First opt	d_h	chunk	c_emb	t_emb	grads / task
CIFAR-10 HN (main)	5	10	50	AdamW	32	6 000	16	16	80
CIFAR-100 HN [†]	10	10	50	AdamW	32	6 000	16	16	80
MNIST HN ($d_h=8$, suffocated) [†]	5	15	15	AdamW	8	64	32	32	100
d_h sweep — $d_h = 4$	5	15	15	AdamW	4	64	32	32	80
d_h sweep — $d_h = 8$	5	15	15	AdamW	8	64	32	32	100
d_h sweep — $d_h = 16$	5	15	15	AdamW	16	64	32	32	80

G.3 Normalization Ablation

All three conditions share the same architecture: Split-CIFAR10, 5 tasks, `max_ep = 50`, `chunk = 6 000`, $d_h = 32$, `batch = 32`, AdamW first-task at $\eta = 10^{-3}$, iFOPNG only.

Table 5: **Normalization ablation conditions (Sub-RQ2)**. The CSVs for conditions 1 and 2 also contain pilot runs at $\text{chunk} = 256$, $d_h = 64$; these are excluded. Only $\text{chunk} = 6000$, $d_h = 32$ runs are used.

Condition	normalize	QR on G	Max-scale $\hat{F}$	Seeds	Accuracy	BWT
No normalization	False	×	×	3	72.2%	−0.070
Gradient-only	True	✓	×	3	68.9%	−0.060
Full normalization	True	✓	✓	5	85.8%	−0.013

G.4 Sub-RQ4 and Compression Ablation

Both Sub-RQ4 conditions and all three compression ablation conditions use 20-task Permuted-MNIST, TargetNetwork, `normalize = False`, `batch = 10`.

Table 6: **Sub-RQ4 accumulation parameters** ($n = 5$ seeds, $\alpha = 0.5$ for EMA).

Operator	Tasks	ep/task	max_ep	LR	λ	Fisher samp.
MAX (iFOPNG)	20	5	5	10^{-4}	10^{-2}	60 000
EMA (iFOPNG_ema, $\alpha = 0.5$)	20	5	5	10^{-4}	10^{-2}	60 000

Table 7: **Compression ablation parameters** ($n = 5$ seeds each, seeds $\{42, 111, 811, 1234, 2137\}$). Note: $\lambda = 10^{-3}$ differs from the Sub-RQ4 baseline ($\lambda = 10^{-2}$); the ablation is internally controlled but not directly comparable to Sub-RQ4 on this hyperparameter.

Strategy	Tasks	ep/task	max_ep	LR	λ	max_dirs
SVD (Exp 421)	20	5	5	10^{-4}	10^{-3}	400
FIFO (Exp 422)	20	5	5	10^{-4}	10^{-3}	400
STOP (Exp 423)	20	5	5	10^{-4}	10^{-3}	400

G.5 SGD Convergence Threshold Diagnostic

The SGD threshold diagnostic (Figure 15) sweeps $\eta \in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$ on Split-MNIST SH (TargetNetwork, 1 task, 5 epochs). Convergence threshold: $\eta = 10^{-4}$. All standalone experiments in this thesis use $\eta_{\text{first}} = 10^{-3}$ except Split-MNIST MH, which follows the Garg et al. (2026) protocol at $\eta_{\text{first}} = 10^{-5}$.

G.6 Notes on Confounds and Uncertainties

EXP 407 `first_task_opt = none`. The run config records `first_task_opt = None` and `first_task_lr = None`. The figure title annotates “AdamW first-task,” likely reflecting a code-level default: Exps 412 and 413 (same architecture, added later) explicitly log `adamw`, suggesting Exp 407 predates that config field.

EXP 411 `PARAMETER HETEROGENEITY`. The CIFAR-100 HN CSV contains 222 runs across a wide range of `chunk_size`, `hyper_hidden_dim`, and normalization settings — an exploratory search prior to settling on the published architecture. The 35 completed runs at `chunk=6000`, $d_h=32$, `normalize=True` are used for the reported results; all other runs are pilots.

`NORMALIZATION ABLATION PILOT RUNS`. Exps 409 and 410 each contain additional runs at `chunk=256`, $d_h=64$ (an earlier pilot) alongside the published `chunk=6000`, $d_h=32$ runs. The pilot runs reach near-random accuracy ($\approx 50\text{--}61\%$) and are excluded; the ablation is architecturally controlled.

`EWC  $\lambda$  IDENTIFICATION`. The `lam` field is shared between EWC’s λ and the projection damping term, so per-method EWC values cannot be extracted from the CSV without row-level method filtering; Table 3 values are inferred from the set of distinct `lam` values per experiment.

`$\lambda$  DISCREPANCY BETWEEN SUB-RQ4 AND COMPRESSION ABLATION`. The Sub-RQ4 experiments (Exps 414–415) use $\lambda = 10^{-2}$, while the compression ablation (Exps 421–423) uses $\lambda = 10^{-3}$. The compression ablation compares SVD, FIFO, and STOP under identical conditions and is internally valid; the comparison against the Sub-RQ4 unconstrained baseline on absolute accuracy is confounded by this difference and should not be made.